

DECLARATION

I hereby declare that the work presented in this thesis entitled **Machine Learning Based Approaches for Modeling Qubit Decoherence** is my original work and has not been submitted elsewhere for the award of any degree, diploma, or fellowship.

Signature of Candidate

Arjyadeb Sengupta

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Quantum computing has emerged as a powerful paradigm, extending computational capabilities beyond classical systems. This shift poses challenges to classical cryptographic protocols, motivating two parallel directions of research. The first focuses on post-quantum cryptography, designing classical algorithms resistant to quantum attacks. The second explores quantum-native approaches such as Quantum Key Distribution (QKD). However, practical deployment of QKD is limited by decoherence in photonic qubits, where interaction with the environment leads to loss of quantum coherence. In this work, we explore whether machine learning can capture and model decoherence from data-driven observations.

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Part I

Foundations

1 Mathematical Aspects required

1.1 References

1. Nielsen, M. A., & Chuang, I. L. (2010). *Quantum Computation and Quantum Information*. Cambridge University Press.
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1.2 Introduction

Quantum computing relies heavily on the mathematical formalism given by linear algebra and functional analysis. This section summarizes the essential mathematical tools required to understand quantum states, quantum operations, and their manipulation within computational frameworks. The focus is on clarity and structural understanding rather than physical interpretation.

1.3 Hilbert Spaces

A Hilbert space $\mathcal{H}$ is a complete inner product space over the field $\mathbb{C}$. The completeness relation ensures that every Cauchy sequence in $\mathcal{H}$ converges to a vector within the space. In quantum mechanics, the state of a physical system is represented by a unit vector in a complex Hilbert space.

For finite-dimensional quantum systems, $\mathcal{H} \cong \mathbb{C}^n$, equipped with the standard inner product

$$\langle \phi | \psi \rangle = \sum_{i=1}^n \phi_i^* \psi_i.$$

1.4 Dirac Notation

Dirac notation provides a compact and expressive way to represent vectors and linear functionals in Hilbert spaces. Vectors are denoted by *kets* $|\psi\rangle$, while their dual vectors are denoted by *bras* $\langle\psi|$.

This notation naturally encodes inner products, outer products, and operator actions in a unified framework.

1.5 Ket Vectors

A ket $|\psi\rangle \in \mathcal{H}$ represents the state of a quantum system. In a chosen orthonormal basis $\{|i\rangle\}$, a ket can be expanded as

$$|\psi\rangle = \sum_i \psi_i |i\rangle,$$

where $\psi_i \in \mathbb{C}$ are probability amplitudes satisfying $\sum_i |\psi_i|^2 = 1$.

1.6 Bra Vectors

The bra corresponding to a ket $|\psi\rangle$ is defined via the Hermitian conjugate:

$$\langle\psi| = (|\psi\rangle)^\dagger.$$

Bras act as linear functionals on kets, producing complex numbers through inner products.

1.7 Outer Products

Given two vectors $|\psi\rangle$ and $|\phi\rangle$, the outer product

$$|\psi\rangle\langle\phi|$$

defines a linear operator on $\mathcal{H}$. Outer products play a central role in constructing projectors, density operators, and measurement operators.

1.8 Linear Operators and Outer Product Decomposition

Any linear operator A acting on a finite-dimensional Hilbert space can be expressed as

$$A = \sum_{i,j} A_{ij} |i\rangle\langle j|,$$

where A_{ij} are the matrix elements of A in the chosen basis. This decomposition is fundamental in quantum circuit design and operator learning.

1.9 Pauli Operators

The Pauli operators form a basis for 2×2 Hermitian operators:

$$\sigma_X = |0\rangle\langle 1| + |1\rangle\langle 0|,$$

$$\sigma_Y = -i|0\rangle\langle 1| + i|1\rangle\langle 0|,$$

$$\sigma_Z = |0\rangle\langle 0| - |1\rangle\langle 1|.$$

They are central to quantum dynamics, error models, and measurement theory.

1.10 Spectral Decomposition

Any Hermitian operator A admits a spectral decomposition

$$A = \sum_k \lambda_k |k\rangle\langle k|,$$

where $\lambda_k \in \mathbb{R}$ are eigenvalues and $|k\rangle$ are orthonormal eigenvectors. Spectral decompositions underpin quantum measurement theory and observable expectation values.

1.11 Tensor Products

1.11.1 Tensor Product of Hilbert Spaces

Composite quantum systems are described using tensor products of Hilbert spaces. If subsystems A and B are associated with $\mathcal{H}_A$ and $\mathcal{H}_B$, the joint system is described by

$$\mathcal{H}_{AB} = \mathcal{H}_A \otimes \mathcal{H}_B.$$

1.11.2 Tensor Product of States

Given

$$|\psi\rangle = \sum_i \alpha_i |i\rangle, \quad |\phi\rangle = \sum_j \beta_j |j\rangle,$$

their joint state is

$$|\psi\rangle \otimes |\phi\rangle = \sum_{i,j} \alpha_i \beta_j |i\rangle \otimes |j\rangle.$$

A state that cannot be factored into such a product is called *entangled*. For example,

$$|\Phi^+\rangle = \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle).$$

1.11.3 Tensor Product of Operators

For operators A on $\mathcal{H}_A$ and B on $\mathcal{H}_B$,

$$(A \otimes B)(|i\rangle \otimes |j\rangle) = (A|i\rangle) \otimes (B|j\rangle).$$

This construction is fundamental for multi-qubit gates and subsystem-local operations.

1.11.4 Tensor Products and Measurement

A measurement acting only on subsystem A is represented as

$$M_A \otimes I_B,$$

where I_B is the identity on $\mathcal{H}_B$.

1.12 Functions of Normal Operators

If an operator A is normal, functions of A can be defined via its spectral decomposition:

$$f(A) = \sum_k f(\lambda_k) |k\rangle \langle k|.$$

This is essential for defining unitary evolutions such as e^{-iHt} .

1.13 Partial Trace

The *partial trace* is an operation that allows me to extract the state of a subsystem from a composite system. Consider a bipartite system with Hilbert space $\mathcal{H}_A \otimes \mathcal{H}_B$ and a density matrix ρ_{AB} . The reduced density matrix of subsystem A is obtained by tracing out subsystem B :

$$\rho_A = \text{Tr}_B[\rho_{AB}]. \quad (1)$$

Matrix Example: Let

$$\rho_{AB} = \begin{pmatrix} 1/2 & 0 & 0 & 1/2 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 1/2 & 0 & 0 & 1/2 \end{pmatrix}, \quad \text{with basis } \{|00\rangle, |01\rangle, |10\rangle, |11\rangle\}. \quad (2)$$

To compute $\rho_A = \text{Tr}_B[\rho_{AB}]$, I sum over the diagonal blocks corresponding to subsystem B :

$$\rho_A = \begin{pmatrix} \rho_{00,00} + \rho_{01,01} & \rho_{00,10} + \rho_{01,11} \\ \rho_{10,00} + \rho_{11,01} & \rho_{10,10} + \rho_{11,11} \end{pmatrix} = \begin{pmatrix} 1/2 + 0 & 0 + 1/2 \\ 0 + 1/2 & 0 + 1/2 \end{pmatrix} = \begin{pmatrix} 1/2 & 1/2 \\ 1/2 & 1/2 \end{pmatrix}.$$

Outer Product Approach: If ρ_{AB} is expressed in the outer product form

$$\rho_{AB} = \sum_{ijkl} \rho_{ij,kl} |i\rangle \langle k| \otimes |j\rangle \langle l|, \quad (3)$$

then the partial trace over B is

$$\text{Tr}_B[\rho_{AB}] = \sum_{ij} \sum_m \rho_{im,jm} |i\rangle \langle j|, \quad (4)$$

where the sum over m runs over an orthonormal basis of $\mathcal{H}_B$.

1.14 Partial Transpose

The *partial transpose* is an operation useful in entanglement detection. For a bipartite density matrix ρ_{AB} , the partial transpose with respect to subsystem B is defined in the outer product form as:

$$\rho_{AB}^{T_B} = \sum_{ijkl} \rho_{ij,kl} |i\rangle \langle k| \otimes (|j\rangle \langle l|)^T = \sum_{ijkl} \rho_{ij,kl} |i\rangle \langle k| \otimes |l\rangle \langle j|. \quad (5)$$

Matrix Example: Take the Bell state density matrix

$$\rho_{AB} = \frac{1}{2} \begin{pmatrix} 1 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 1 \end{pmatrix}. \quad (6)$$

The partial transpose w.r.t subsystem B swaps the B indices in the matrix elements:

$$\rho_{AB}^{T_B} = \frac{1}{2} \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}.$$

Outer Product Approach: Using the basis expansion

$$\rho_{AB} = \sum_{ijkl} \rho_{ij,kl} |i\rangle\langle k| \otimes |j\rangle\langle l|,$$

the partial transpose on B gives

$$\rho_{AB}^{T_B} = \sum_{ijkl} \rho_{ij,kl} |i\rangle\langle k| \otimes |l\rangle\langle j|,$$

which is equivalent to swapping the subsystem B indices in each outer product.

Summary of Procedures:

1. **Partial Trace:** Sum over the indices of the subsystem being traced out.
2. **Partial Transpose:** Swap the row and column indices of the chosen subsystem in each matrix element.
3. Both operations can be implemented directly in matrix form or using the outer product expansion of the density matrix.

1.15 Conclusion

These mathematical tools form the backbone of quantum computing. They provide a rigorous framework for understanding quantum states, operations, measurements, and learning architectures built on quantum data representations.

2 Physics Concepts Needed

2.1 References

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2.2 Postulates of Quantum Mechanics

2.2.1 Postulate 1: State of a Quantum System

A quantum system is completely described by a state vector $|\psi\rangle$ belonging to a complex Hilbert space $\mathcal{H}$. In position representation, the state is described by a wave function

$$\psi(\mathbf{x}, t) = \langle \mathbf{x} | \psi(t) \rangle,$$

which satisfies the normalization condition

$$\int |\psi(\mathbf{x}, t)|^2 d\mathbf{x} = 1.$$

2.2.2 Postulate 2: Observables as Operators

Every physical observable A is represented by a linear Hermitian operator $\hat{A}$ acting on $\mathcal{H}$. The possible outcomes of a measurement correspond to the eigenvalues of $\hat{A}$:

$$\hat{A} |a_n\rangle = a_n |a_n\rangle.$$

2.2.3 Postulate 3: Measurement and Expectation Values

The expectation value of an observable $\hat{A}$ in the state $|\psi\rangle$ is given by

$$\langle A \rangle = \langle \psi | \hat{A} | \psi \rangle.$$

The probability of obtaining outcome a_n is

$$p(a_n) = |\langle a_n | \psi \rangle|^2.$$

2.3 Wave Function and Schrödinger Equation

2.3.1 Schrödinger Equation

The time evolution of a closed quantum system is governed by

$$i\hbar \frac{d}{dt} |\psi(t)\rangle = \hat{H} |\psi(t)\rangle,$$

where $\hat{H}$ is the Hamiltonian operator.

In position space,

$$i\hbar \frac{\partial \psi(\mathbf{x}, t)}{\partial t} = \left(-\frac{\hbar^2}{2m} \nabla^2 + V(\mathbf{x}) \right) \psi(\mathbf{x}, t).$$

The formal solution is

$$|\psi(t)\rangle = e^{-i\hat{H}t/\hbar} |\psi(0)\rangle.$$

2.4 Quantum Harmonic Oscillator

The quantum harmonic oscillator (QHO) is a fundamental system with Hamiltonian

$$\hat{H} = \frac{\hat{p}^2}{2m} + \frac{1}{2} m \omega^2 \hat{x}^2,$$

where $\hat{x}$ and $\hat{p}$ are position and momentum operators, m is the mass, and ω is the angular frequency.

2.4.1 Ladder Operators

Introducing the ladder (creation and annihilation) operators:

$$\hat{a} = \frac{1}{\sqrt{2\hbar m \omega}} (m\omega \hat{x} + i\hat{p}), \quad \hat{a}^\dagger = \frac{1}{\sqrt{2\hbar m \omega}} (m\omega \hat{x} - i\hat{p}),$$

the Hamiltonian becomes

$$\hat{H} = \hbar \omega \left(\hat{a}^\dagger \hat{a} + \frac{1}{2} \right).$$

2.4.2 Energy Eigenstates

The eigenstates of $\hat{H}$ are number states $|n\rangle$:

$$\hat{H} |n\rangle = \hbar \omega \left(n + \frac{1}{2} \right) |n\rangle, \quad n = 0, 1, 2, \dots$$

These states have well-defined energies but indefinite position and momentum.

2.4.3 Coherent States and Minimum Uncertainty

A coherent state $|\alpha\rangle$ is defined as an eigenstate of the annihilation operator:

$$\hat{a}|\alpha\rangle = \alpha|\alpha\rangle, \quad \alpha \in \mathbb{C}.$$

Coherent states are physically significant because they:

- Minimize the Heisenberg uncertainty relation: $\Delta x \Delta p = \hbar/2$.
- Exhibit classical-like behavior: the expectation values $\langle \hat{x}(t) \rangle$ and $\langle \hat{p}(t) \rangle$ follow the classical harmonic oscillator equations.
- Form an overcomplete basis, allowing decomposition of arbitrary states.

Mathematically, a coherent state can be expressed as a superposition of number states:

$$|\alpha\rangle = e^{-|\alpha|^2/2} \sum_{n=0}^{\infty} \frac{\alpha^n}{\sqrt{n!}} |n\rangle.$$

These states are widely used in quantum optics and laser physics as they bridge quantum and classical descriptions.

2.5 Measurement Theory

2.5.1 Projective Measurements

Projective measurements are described by orthogonal projectors

$$P_i = |i\rangle\langle i|, \quad \sum_i P_i = I.$$

The probability of outcome i is

$$p(i) = \langle \psi | P_i | \psi \rangle.$$

2.5.2 Positive Operator-Valued Measures (POVMs)

Generalized measurements are described by POVM elements $\{E_i\}$ such that

$$E_i \geq 0, \quad \sum_i E_i = I.$$

For a density operator ρ ,

$$p(i) = \text{Tr}(\rho E_i).$$

This is of extreme importance since quantum measurement causes the quantum state of a system to change. In particular, when we will get to the practical execution of QKD, this will turn out to be one of its biggest strengths.

2.6 Density Operators

2.6.1 Mathematical Definition and Core Properties

In quantum mechanics, a system whose state is completely known is described by a single state vector $|\psi\rangle$ (a pure state). When a system exhibits statistical or classical uncertainty, it is described as a mixed state—an ensemble of pure states $|\psi_i\rangle$, each occurring with a classical probability p_i .

The density operator (or density matrix) ρ is defined as:

$$\rho = \sum_i p_i |\psi_i\rangle\langle\psi_i| \tag{7}$$

where the classical probabilities satisfy $p_i \geq 0$ and $\sum_i p_i = 1$. For a pure state, the expression simplifies to:

$$\rho = |\psi\rangle\langle\psi| \tag{8}$$

An operator ρ acting on a Hilbert space $\mathcal{H}$ is a valid density operator if and only if it satisfies three essential conditions:

1. **Hermiticity:** The operator is self-adjoint, ensuring real-valued physical observables.

$$\rho^\dagger = \rho \quad (9)$$

2. **Unit Trace:** The sum of the probabilities of all mutually exclusive states equals 1.

$$\text{Tr}(\rho) = 1 \quad (10)$$

3. **Positive Semi-definiteness:** For any vector $|\phi\rangle \in \mathcal{H}$, the expectation value is non-negative, preventing negative probabilities.

$$\langle \phi | \rho | \phi \rangle \geq 0 \quad (11)$$

2.6.2 Distinguishing Pure and Mixed States

The purity of a quantum state is verified by computing the trace of the square of the density operator:

$$\text{Tr}(\rho^2) \leq 1 \quad (12)$$

- **Pure State:** $\text{Tr}(\rho^2) = 1$ (since $\rho^2 = \rho$).
- **Mixed State:** $\text{Tr}(\rho^2) < 1$. For a d -dimensional Hilbert space, the lower bound is $\frac{1}{d}$, which corresponds to the maximally mixed state.

2.6.3 Concrete Matrix Examples

Consider a two-level system (qubit) defined in the computational basis $\{|0\rangle, |1\rangle\}$, where:

$$|0\rangle = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \quad |1\rangle = \begin{pmatrix} 0 \\ 1 \end{pmatrix} \quad (13)$$

Example 1: Pure State Let the system be in a coherent superposition state $|\psi\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)$. The density matrix is:

$$\rho_{\text{pure}} = |\psi\rangle\langle\psi| = \frac{1}{2} \begin{pmatrix} 1 \\ 1 \end{pmatrix} \begin{pmatrix} 1 & 1 \end{pmatrix} = \begin{pmatrix} \frac{1}{2} & \frac{1}{2} \\ \frac{1}{2} & \frac{1}{2} \end{pmatrix} \quad (14)$$

This matrix satisfies $\text{Tr}(\rho_{\text{pure}}) = \frac{1}{2} + \frac{1}{2} = 1$ and $\text{Tr}(\rho_{\text{pure}}^2) = 1$.

Example 2: Mixed State Let the system be an ensemble where there is a 50% classical probability of being in state $|0\rangle$ and a 50% classical probability of being in state $|1\rangle$. The density matrix contains no phase relation:

$$\rho_{\text{mixed}} = \frac{1}{2}|0\rangle\langle 0| + \frac{1}{2}|1\rangle\langle 1| = \frac{1}{2} \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix} + \frac{1}{2} \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} \frac{1}{2} & 0 \\ 0 & \frac{1}{2} \end{pmatrix} \quad (15)$$

This matrix satisfies $\text{Tr}(\rho_{\text{mixed}}) = \frac{1}{2} + \frac{1}{2} = 1$ and $\text{Tr}(\rho_{\text{mixed}}^2) = \frac{1}{4} + \frac{1}{4} = \frac{1}{2} < 1$.

2.6.4 Population Probability

The diagonal elements of the density matrix, ρ_{ii} , represent the **population probabilities**. Given an orthonormal basis $\{|i\rangle\}$, ρ_{ii} is the probability of finding the system in the specific basis state $|i\rangle$ upon measurement:

$$\rho_{ii} = \langle i | \rho | i \rangle \quad (16)$$

In both matrix examples provided above, the populations are $\rho_{00} = \frac{1}{2}$ and $\rho_{11} = \frac{1}{2}$, meaning a measurement in the computational basis yields $|0\rangle$ or $|1\rangle$ with equal likelihood.

2.6.5 Coherence

The off-diagonal elements, ρ_{ij} (where $i \neq j$), define quantum **coherence**. They quantify the physical interference and phase relationships between the basis states $|i\rangle$ and $|j\rangle$:

$$\rho_{ij} = \langle i | \rho | j \rangle \quad (i \neq j) \quad (17)$$

- In ρ_{pure} , $\rho_{01} = \rho_{10} = \frac{1}{2}$, indicating maximum quantum coherence between $|0\rangle$ and $|1\rangle$.
- In ρ_{mixed} , $\rho_{01} = \rho_{10} = 0$, meaning all quantum coherence has been destroyed, leaving only a classical statistical mixture.

2.6.6 Survival Probability

The **survival probability** $P(t)$ is the probability that a quantum system, initially prepared in a state $\rho(0)$ at time $t = 0$, remains in that exact state at a later time t . It is calculated using the trace operation:

$$P(t) = \text{Tr}(\rho(t)\rho(0)) \quad (18)$$

If the initial state is pure, $\rho(0) = |\psi(0)\rangle\langle\psi(0)|$, the expression simplifies to the square of the overlap amplitude:

$$P(t) = \langle\psi(0)|\rho(t)|\psi(0)\rangle = |\langle\psi(0)|\psi(t)\rangle|^2 \quad (19)$$

2.6.7 Physical Importance and Dynamics

The density operator framework is mathematically required to describe several core physical phenomena:

- **Expectation Values:** The expectation value of any linear observable represented by an operator A is given by:

$$\langle A \rangle = \text{Tr}(\rho A) \quad (20)$$

- **Description of Subsystems (Partial Trace):** For a composite system described by a joint density matrix ρ_{AB} , the local state of subsystem A is obtained by evaluating the partial trace over the degrees of freedom of subsystem B :

$$\rho_A = \text{Tr}_B(\rho_{AB}) \quad (21)$$

If ρ_{AB} is an entangled pure state, ρ_A is strictly a mixed state.

- **Time Evolution (Von Neumann Equation):** The temporal dynamics of a density operator under a deterministic Hamiltonian H is governed by:

$$i\hbar \frac{\partial \rho(t)}{\partial t} = [H, \rho(t)] \quad (22)$$

where $[H, \rho(t)] = H\rho(t) - \rho(t)H$ is the commutator.

- **Quantum Decoherence:** When a system interacts with an external environment, the off-diagonal terms ρ_{ij} ($i \neq j$) decay over time toward zero. This physical transformation from ρ_{pure} to ρ_{mixed} describes how quantum superpositions degrade into classical probability distributions.

2.7 Commutators and Uncertainty Relations

$$[\hat{A}, \hat{B}] = \hat{A}\hat{B} - \hat{B}\hat{A}.$$

$$\Delta A \Delta B \geq \frac{1}{2} |\langle [\hat{A}, \hat{B}] \rangle|.$$

$$[\hat{x}, \hat{p}] = i\hbar.$$

2.8 Wave–Particle Duality and Foundational Experiments

2.8.1 Young’s Double Slit Experiment (Detailed)

Consider a source emitting particles (photons, electrons, or even larger molecules) toward a barrier with two narrow slits. Beyond the slits is a detection screen that records where particles arrive.

Let $\psi_1(\mathbf{x})$ denote the wave function if the particle passes through slit 1, and $\psi_2(\mathbf{x})$ if it passes through slit 2. When both slits are open, quantum mechanics dictates that the system is described by a coherent superposition:

$$\psi(\mathbf{x}) = \psi_1(\mathbf{x}) + \psi_2(\mathbf{x}).$$

The probability of detecting the particle at position $\mathbf{x}$ on the screen is

$$P(\mathbf{x}) = |\psi(\mathbf{x})|^2 = |\psi_1(\mathbf{x}) + \psi_2(\mathbf{x})|^2.$$

Expanding gives

$$P(\mathbf{x}) = |\psi_1(\mathbf{x})|^2 + |\psi_2(\mathbf{x})|^2 + 2 \text{Re}(\psi_1^*(\mathbf{x})\psi_2(\mathbf{x})),$$

where the last term represents interference from the superposition. This interference term is a direct manifestation of the wave-like nature of particles and cannot be explained classically.

Observation and Measurement: If a measurement device is placed to detect which slit the particle passes through, coherence is destroyed. The interference term vanishes, and the probability distribution reduces to the classical sum $|\psi_1|^2 + |\psi_2|^2$. This demonstrates that the act of measurement affects the system—an essential principle of quantum mechanics.

2.9 Polarization and Optical Elements

Polarization states of photons are represented as qubit states:

$$|H\rangle = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \quad |V\rangle = \begin{pmatrix} 0 \\ 1 \end{pmatrix}.$$

Polarizing Beam Splitter (PBS) and Pockels Cell:

- A PBS separates horizontal and vertical polarization components. By routing these components along different paths, it allows preparation and measurement in specific bases.
- A Pockels cell is an electro-optic device that rotates the polarization state depending on applied voltage. It can transform $|H\rangle$ and $|V\rangle$ into arbitrary superpositions like $\alpha|H\rangle + \beta|V\rangle$.
- By combining PBS and Pockels cells, one can prepare any desired qubit state, implement basis changes for measurement, or create entangled photon pairs.

Malus's law governs intensity after polarizers:

$$I = I_0 \cos^2 \theta.$$

2.10 Quantum Superposition, Entanglement, and Teleportation

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle.$$

2.10.1 Bell States

$$|\Phi^\pm\rangle = \frac{1}{\sqrt{2}}(|00\rangle \pm |11\rangle), \quad |\Psi^\pm\rangle = \frac{1}{\sqrt{2}}(|01\rangle \pm |10\rangle).$$

2.10.2 Quantum Teleportation (Detailed)

Quantum teleportation is a protocol to transmit an unknown qubit state $|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$ from Alice to Bob using:

1. A shared entangled pair (e.g., $|\Phi^+\rangle_{AB}$).
2. Classical communication of measurement results.

Step 1: Initial Setup Alice has the state $|\psi\rangle_C$ to teleport. Alice and Bob share a Bell pair $|\Phi^+\rangle_{AB} = \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$. The total state is

$$|\Psi\rangle_{CAB} = |\psi\rangle_C \otimes |\Phi^+\rangle_{AB}.$$

Step 2: Bell Measurement by Alice Alice performs a Bell basis measurement on her qubits C and A . This projects the system onto one of four Bell states:

$$|\Phi^\pm\rangle_{CA}, \quad |\Psi^\pm\rangle_{CA}.$$

Step 3: Classical Communication Alice sends the 2-bit result of her measurement to Bob.

Step 4: Conditional Operations by Bob Depending on Alice's outcome, Bob applies a corresponding unitary:

$$\text{Outcome} \Rightarrow \text{Bob applies } I, X, Z, XZ$$

where X and Z are Pauli operators. After this, Bob's qubit B is in the exact state $|\psi\rangle$.

Key Points:

- No physical transfer of the qubit occurs; only classical bits and entanglement are used.
- Original state at Alice's side is destroyed by measurement, satisfying the no-cloning theorem.
- Teleportation relies on superposition, entanglement, and coherent operations.

2.11 Pure and Mixed States

$$\text{Tr}(\rho^2) = 1 \quad (\text{pure}), \quad \text{Tr}(\rho^2) < 1 \quad (\text{mixed}).$$

2.12 Coherent Superposition of Quantum States

If $|\phi_1\rangle$ and $|\phi_2\rangle$ are valid states, then

$$|\psi\rangle = \alpha |\phi_1\rangle + \beta |\phi_2\rangle$$

is also a valid state.

2.12.1 Relative Phase and Interference

Only the relative phase between coefficients affects physical predictions and leads to interference.

2.12.2 Superposition vs Classical Mixtures

Off-diagonal density matrix elements distinguish coherent superpositions from classical mixtures.

2.12.3 Measurement and Decoherence

Measurement and environmental interaction destroy coherence, converting pure states into mixed states.

2.12.4 Relevance to Quantum Cryptography

Security in quantum cryptographic protocols arises from the impossibility of measuring quantum states without disturbing their coherence. This is the physical principle that enforces security: any attempt to eavesdrop introduces errors detectable by legitimate parties. Coherent superpositions of non-orthogonal states are exploited in protocols such as BB84 and B92 to ensure secure key distribution.

3 Quantum Measure**3.1 References****5.1 References**

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2. Hayashi, M. (2017). *Quantum Information Theory: Mathematical Foundation*. Springer.
3. Petz, D. (2008). *Quantum Information Theory and Quantum Statistics*. Springer.
4. Nielsen, M. A., & Chuang, I. L. (2010). *Quantum Computation and Quantum Information*. Cambridge University Press.

Trace Distance

The trace distance between two density operators ρ and σ acting on a complex Hilbert space $\mathcal{H}$ is a metric defined via the trace norm as:

$$D(\rho, \sigma) = \frac{1}{2} \|\rho - \sigma\|_1 = \frac{1}{2} \text{Tr} \sqrt{(\rho - \sigma)^\dagger (\rho - \sigma)}$$

where $|A| = \sqrt{A^\dagger A}$ is the unique positive square root of $A^\dagger A$. Since the difference operator $\rho - \sigma$ is self-adjoint, $|\rho - \sigma|$ corresponds to the absolute values of its eigenvalues. If $\lambda_i \in \mathbb{R}$ are the eigenvalues of the operator $(\rho - \sigma)$, the definition simplifies to:

$$D(\rho, \sigma) = \frac{1}{2} \sum_i |\lambda_i|$$

Properties:

- **Bounds:** $0 \leq D(\rho, \sigma) \leq 1$. $D(\rho, \sigma) = 0$ if and only if $\rho = \sigma$, and $D(\rho, \sigma) = 1$ if and only if ρ and σ have orthogonal supports ($\text{Tr}(\rho\sigma) = 0$).
- **Operational Interpretation:** It represents the maximum probability of distinguishing the two states in a single measurement setup (maximized over all POVM elements $0 \leq P \leq I$):

$$P_{\max} = \frac{1}{2} (1 + D(\rho, \sigma))$$

Fidelity

The fidelity between two density operators ρ and σ is a measure of their similarity, defined via the Uhlmann-Jozsa convention as:

$$F(\rho, \sigma) = \left(\text{Tr} \sqrt{\sqrt{\rho} \sigma \sqrt{\rho}} \right)^2$$

Note on Convention: In certain literature (such as Nielsen and Chuang), fidelity is defined without the square, as $\sqrt{F(\rho, \sigma)} = \text{Tr} \sqrt{\sqrt{\rho} \sigma \sqrt{\rho}}$.

Pure State Simplification:

If $\rho = |\psi\rangle\langle\psi|$ is a pure state (a rank-1 projection operator), the expression collapses to the operational overlap:

$$F(|\psi\rangle\langle\psi|, \sigma) = \langle\psi|\sigma|\psi\rangle$$

If both states are pure ($\rho = |\psi\rangle\langle\psi|$ and $\sigma = |\phi\rangle\langle\phi|$):

$$F(|\psi\rangle\langle\psi|, |\phi\rangle\langle\phi|) = |\langle\psi|\phi\rangle|^2$$

Properties:

- **Bounds:** $0 \leq F(\rho, \sigma) \leq 1$. $F(\rho, \sigma) = 1$ if and only if $\rho = \sigma$, and $F(\rho, \sigma) = 0$ if and only if the states are completely orthogonal ($\text{Tr}(\rho\sigma) = 0$).
- **Symmetry:** $F(\rho, \sigma) = F(\sigma, \rho)$.

Relations Between Trace Distance and Fidelity

The trace distance and fidelity constrain each other through the Fuchs-van de Graaf inequalities:

$$1 - \sqrt{F(\rho, \sigma)} \leq D(\rho, \sigma) \leq \sqrt{1 - F(\rho, \sigma)}$$

4 Introductory Quantum Computing

4.1 References

1. Nielsen, M. A., & Chuang, I. L. (2010). *Quantum Computation and Quantum Information*. Cambridge University Press.
2. Kaye, P., Laflamme, R., & Mosca, M. (2007). *An Introduction to Quantum Computing*. Oxford University Press.
3. Mermin, N. D. (2007). *Quantum Computer Science: An Introduction*. Cambridge University Press.
4. Hirvensalo, M. (2001). *Quantum Computing*. Springer.

4.2 Definition of a Quantum Bit

A quantum bit seen through the lenses of computer science, is the smallest unit of a quantum computer. The physics perspective of a quantum bit(henceforward, I will call it qubit) is much richer and illuminating. It is any physical system having two distinguishable quantum states. You already saw one example, the photon, which was in a superposition of 2 polarization angles. Another example would be a quantum particle bound between 2 distinguishable energy levels. Mathematically, we can see it as an element of 2 dimensional complex Hilbert Space. The physical definition allows us to construct or manifest qubits into existence, the mathematical definition lets us perform pen and paper calculations, and the computer science definition allows us to turn it into daily life objects.

4.3 Physical ways to construct qubits

4.3.1 Photonic Qubits (Linear Optics)

Photons are the most practical medium for Quantum Key Distribution (QKD) due to their ability to travel long distances through vacuum or fiber optics.

- **Polarization Encoding:** Information is encoded in the orientation of the electric field (e.g., $|\rightarrow\rangle, |\uparrow\rangle, |\nearrow\rangle, |\nwarrow\rangle$).
- **Phase Encoding:** Qubits are defined by the relative phase difference between two paths in an interferometer.
- **Time-Bin Encoding:** Information is stored based on the arrival time of a photon (e.g, "early" vs. "late" bins).

4.3.2 Superconducting Transmon Qubits

These artificial atoms rely on the Josephson Effect to create non-linear oscillators.

- **Mechanism:** A superconducting circuit containing a Josephson junction creates non-equidistant energy levels.
- **Control:** The two lowest energy levels ($|0\rangle$ and $|1\rangle$) are isolated and manipulated using microwave pulses.

4.3.3 Trapped Ion Qubits

Individual atoms are ionized and suspended in a vacuum using electromagnetic Paul traps.

- **Hyperfine Transitions:** Qubits are defined by stable internal electronic states, such as the ground states of $^{171}\text{Yb}^+$.
- **Interaction:** Lasers are used for cooling and to perform multi-qubit gate operations through collective vibrational motion (phonons).

4.3.4 Spin Qubits (Semiconductor Quantum Dots)

This method involves trapping single electrons within semiconductor structures, typically silicon or GaAs.

- **Mechanism:** The qubit is defined by the electron spin (Up or Down) in a magnetic field (Zeeman effect).
- **Advantage:** This approach leverages existing CMOS manufacturing technology for potential scalability.

4.3.5 Neutral Atom Qubits

Neutral atoms are held in place by "optical tweezers" created by focused laser beams.

- **Rydberg States:** High-energy "Rydberg states" allow for strong interactions between atoms via the Rydberg blockade effect.
- **Scalability:** Large 2D or 3D arrays of atoms can be trapped in optical lattices.

4.3.6 Comparison for QKD Implementation

In the context of QKD, photons are the preferred implementation because they are uniquely suited for transmission over long-distance channels. However, these channels are sensitive to noise and photon loss, which are primary factors in calculating the Quantum Bit Error Rate (QBER).

4.4 Bloch Sphere Representation of a qubit

Consider a single qubit. Let $\mathcal{H}_2$ denote the 2D Hilbert space, where $\{|0\rangle, |1\rangle\}$ is an orthonormal basis (onb) of $\mathcal{H}_2$. Any state $|\psi\rangle \in \mathcal{H}_2$ can be written in terms of these basis elements. In particular, for a qubit:

$$|\psi\rangle = a|0\rangle + b|1\rangle, \quad a, b \in \mathbb{C} \quad (23)$$

The normalization condition requires:

$$|a|^2 + |b|^2 = 1 \quad (1) \quad (24)$$

Now, since $a, b \in \mathbb{C}$:

- $a = r_a e^{i\theta_a}, \quad r_a > 0$
- $b = r_b e^{i\theta_b}, \quad r_b > 0$

Substituting these into the state equation:

$$\begin{aligned} |\psi\rangle &= r_a e^{i\theta_a} |0\rangle + r_b e^{i\theta_b} |1\rangle \\ &= r_a e^{i\theta_a} \left(|0\rangle + \frac{r_b}{r_a} e^{i(\theta_b - \theta_a)} |1\rangle \right) \\ &= e^{i\theta_a} (r_a |0\rangle + r_b e^{i\phi} |1\rangle) \end{aligned}$$

where $\phi = \theta_b - \theta_a$.

From equation (1), $r_a^2 + r_b^2 = 1$. We may then parametrize these as:

- $r_a = \cos \frac{\theta}{2}, \quad \theta \in [0, \pi]$
- $r_b = \sin \frac{\theta}{2}$

Since we always deal with normalized states, the global phase ($e^{i\theta_a}$) may be ignored. Thus:

$$|\psi\rangle = \cos \frac{\theta}{2} |0\rangle + \sin \frac{\theta}{2} e^{i\phi} |1\rangle \quad (25)$$

where $\theta \in [0, \pi]$ and $\phi \in [0, 2\pi]$. We may see these as points on a sphere, where $|\psi\rangle \equiv (\theta, \phi)$.

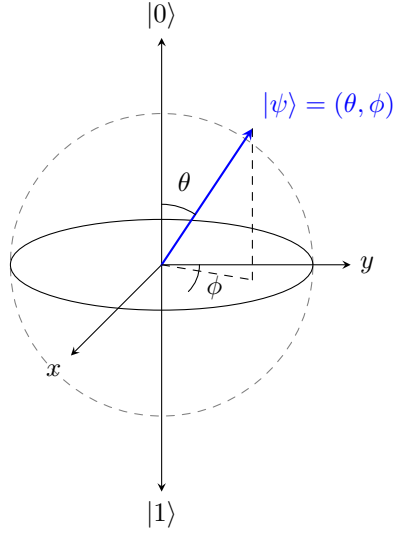


Figure 1: Bloch Sphere Representation

4.5 Multiple Qubit

Multiple qubits can be constructed from single qubits via tensor product.

E.g. Take

- $H_A : \{|0_A\rangle, |1_A\rangle\}$
- $H_B : \{|0_B\rangle, |1_B\rangle\}$

We have,

$$H_A \otimes H_B$$

gives us a **composite space** of qubits.

As a general case, If I have n qubits, I can get

$$H_{2^n} = H_2 \otimes H_2 \otimes \cdots \otimes H_2$$

4.6 Classical Logic Gates

A classical logic gate is a function $f : \{0, 1\}^n \rightarrow \{0, 1\}$, where n is the number of input bits. These are the primary gates:

- **NOT gate:** The NOT gate is a function of a single variable that maps an input to its logical complement.

$$f_{\text{NOT}}(A) = 1 - A$$

If $A = 0, f(0) = 1$ If $A = 1, f(1) = 0$

- **AND Gate:** The AND gate acts like a standard algebraic multiplication. The output is 1 only if both inputs are 1.

$$f_{\text{AND}}(A, B) = A \cdot B$$

This function only returns 1 when $A = 1$ and $B = 1$.

- **OR gate:** The OR gate functions as a logical sum. In binary terms, it is 1 if the sum is 1 or greater.

$$f_{\text{OR}}(A, B) = A + B - (A \cdot B)$$

Note: The $-(A \cdot B)$ term ensures that if both A and B are 1, the result remains 1 (saturating at 1) rather than 2.

- **XOR gate:** The XOR gate is mathematically equivalent to addition modulo 2. This is particularly relevant for us because XOR is used to compare bit strings in QKD protocols to detect eavesdropping.

$$f_{\text{XOR}}(A, B) = A \oplus B = (A + B) \pmod{2}$$

Alternatively, it can be written as: $f_{\text{XOR}}(A, B) = A(1 - B) + B(1 - A)$.

- **NAND gate:** Since NAND is simply NOT-AND, its function is the complement of the product.

$$f_{\text{NAND}}(A, B) = 1 - (A \cdot B)$$

Now, we say a set of classical logic gates is universal if given any other gate, we can construct it using the gates present in the set. I will extend this to quantum logic gates, and give a comparative table.

4.7 Mathematical Construction of Quantum Logic Gates

In the classical case, the gates were Boolean functions. Now if we look in the quantum case, we are essentially operating on a linear combination of $|0\rangle$ and $|1\rangle$. They happen to be elements of a Hilbert space(H). So we need our gate to be an operator on H . Also, observe that the gate transforms a state. So for closed quantum systems, we have an associated unitary transformation. So keeping in mind that we are dealing with a closed quantum system, the quantum logic gate has to be a unitary operator. If we consider an open quantum system, things get a little complicated, but I will discuss it later in my ML QKD program.

4.8 Single Qubit Gates

4.8.1 Characterizing Single Qubit Gates

If we recall the context, in which we defined the Bloch sphere, every qubit on the Bloch sphere is a member of a 2-dimensional complex Hilbert space. So with respect to the basis representation, our single-qubit gate has to be a 2 by 2 unitary matrix. Now there is a special way to represent these matrices in terms of some fixed matrices, which we will see soon.

4.8.2 Quantum X gate

In the computational basis $\{|0\rangle, |1\rangle\}$, the X gate is represented by the following 2×2 unitary matrix:

$$X = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$$

. When applied to a state vector, the X gate swaps the coefficients of $|0\rangle$ and $|1\rangle$: Acting on $|0\rangle$:

$$X|0\rangle = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} 1 \\ 0 \end{pmatrix} = \begin{pmatrix} 0 \\ 1 \end{pmatrix} = |1\rangle$$

Acting on $|1\rangle$:

$$X|1\rangle = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} 0 \\ 1 \end{pmatrix} = \begin{pmatrix} 1 \\ 0 \end{pmatrix} = |0\rangle$$

.

4.8.3 Quantum Y gate

In the computational basis $\{|0\rangle, |1\rangle\}$, the Y gate is defined by the following matrix:

$$Y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}$$

.When applied to the standard basis states, the Y gate flips the state and introduces a complex phase: Acting on $|0\rangle$:

$$Y|0\rangle = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix} \begin{pmatrix} 1 \\ 0 \end{pmatrix} = \begin{pmatrix} 0 \\ i \end{pmatrix} = i|1\rangle$$

Acting on $|1\rangle$:

$$Y|1\rangle = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix} \begin{pmatrix} 0 \\ 1 \end{pmatrix} = \begin{pmatrix} -i \\ 0 \end{pmatrix} = -i|0\rangle$$

.

4.8.4 Quantum Z gate

In the computational basis $\{|0\rangle, |1\rangle\}$, the Z gate is represented by the following diagonal matrix:

$$Z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$$

The Z gate is unique because it does nothing to the $|0\rangle$ state, but flips the sign (phase) of the $|1\rangle$ state: Acting on $|0\rangle$:

$$Z|0\rangle = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \begin{pmatrix} 1 \\ 0 \end{pmatrix} = \begin{pmatrix} 1 \\ 0 \end{pmatrix} = |0\rangle$$

Acting on $|1\rangle$:

$$Z|1\rangle = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \begin{pmatrix} 0 \\ 1 \end{pmatrix} = \begin{pmatrix} 0 \\ -1 \end{pmatrix} = -|1\rangle$$

.

4.8.5 Hadamard Gate

The Hadamard gate (H) is perhaps the most important single-qubit gate in quantum information. It is the gate that creates superposition, moving a qubit from a definite state ($|0\rangle$ or $|1\rangle$) into a state where it is both 0 and 1 simultaneously with equal probability. The Hadamard gate is represented by the following matrix:

$$H = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$$

The H gate maps the computational basis (Z-basis) to the diagonal basis (X-basis): From $|0\rangle$ to "Plus":

$$H|0\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle) = |+\rangle$$

From $|1\rangle$ to "Minus":

$$H|1\rangle = \frac{1}{\sqrt{2}}(|0\rangle - |1\rangle) = |-\rangle$$

Crucially, because the H gate is its own inverse ($H^2 = I$), it also maps the X-basis back to the Z-basis: $H|+\rangle = |0\rangle$, $H|-\rangle = |1\rangle$.

4.9 2 Qubit Gates

If you recall how we constructed multiple-qubit states via tensor product, we had $|a\rangle \otimes |b\rangle$ where $|a\rangle \in \{|0_A\rangle, |1_A\rangle\}$ and $|b\rangle \in \{|0_B\rangle, |1_B\rangle\}$. These gates deal with the if-else conditions. So by construction, the first $|a\rangle$ is designated as the control bit, while $|b\rangle$ is designated as the target bit.

4.9.1 CNOT gate

The CNOT (Controlled-NOT) gate is a fundamental two-qubit gate. It is the primary tool for creating entanglement, which is the "spooky action" at the heart of the project on QKD protocols. The CNOT gate acts on two qubits: a control qubit and a target qubit. If the control qubit is $|0\rangle$, the target qubit remains unchanged. If the control qubit is $|1\rangle$, the target qubit is flipped (an X gate is applied). In the 4-dimensional basis $\{|00\rangle, |01\rangle, |10\rangle, |11\rangle\}$, the CNOT matrix is:

$$CNOT = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{pmatrix}$$

You can think of it as addition modulo 2 on the target bit: $|c, t\rangle \rightarrow |c, c \oplus t\rangle$. $CNOT|00\rangle = |00\rangle$, $CNOT|01\rangle = |01\rangle$, $CNOT|10\rangle = |11\rangle$, $CNOT|11\rangle = |10\rangle$. This is why the CNOT is a "universal" component. If we put the control qubit into superposition first using a Hadamard gate, we create a Bell state (entanglement):

$$CNOT(H \otimes I)|00\rangle = CNOT\left(\frac{|0\rangle + |1\rangle}{\sqrt{2}} \otimes |0\rangle\right) = \frac{|00\rangle + |11\rangle}{\sqrt{2}}$$

.

4.9.2 Controlled U Gate

The Controlled-U (C-U) gate is the generalized version of the CNOT gate. While a CNOT applies a specific X gate to the target, a Controlled-U applies an arbitrary unitary operation U to the target qubit, conditioned on the control qubit being in the state $|1\rangle$. If the control qubit is $|c\rangle$ and the target qubit is $|t\rangle$: If $|c\rangle = |0\rangle$, the state remains $|0\rangle \otimes |t\rangle$. If $|c\rangle = |1\rangle$, the state becomes $|1\rangle \otimes U|t\rangle$. In compact notation:

$$C-U = |0\rangle\langle 0| \otimes I + |1\rangle\langle 1| \otimes U$$

. In the computational basis $\{|00\rangle, |01\rangle, |10\rangle, |11\rangle\}$, the matrix looks like this:

$$C-U = \left(\begin{array}{cc|cc} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ \hline 0 & 0 & u_{11} & u_{12} \\ 0 & 0 & u_{21} & u_{22} \end{array} \right)$$

Table 1: Comparison of Classical and Quantum Logic Gates

Classical Gate	Quantum Gate	Notes on Quantum Logic
NOT	X	Standard bit-flip; preserves unitarity.
XOR	CNOT	Implements $A \oplus B$ while maintaining reversibility.
AND	Toffoli (CCNOT)	Requires an extra qubit to remain reversible.
(N/A)	Hadamard (H)	Generates superposition; essential for BB84 bases.
(N/A)	Pauli-Z	Manages relative phase; critical for QKD security.

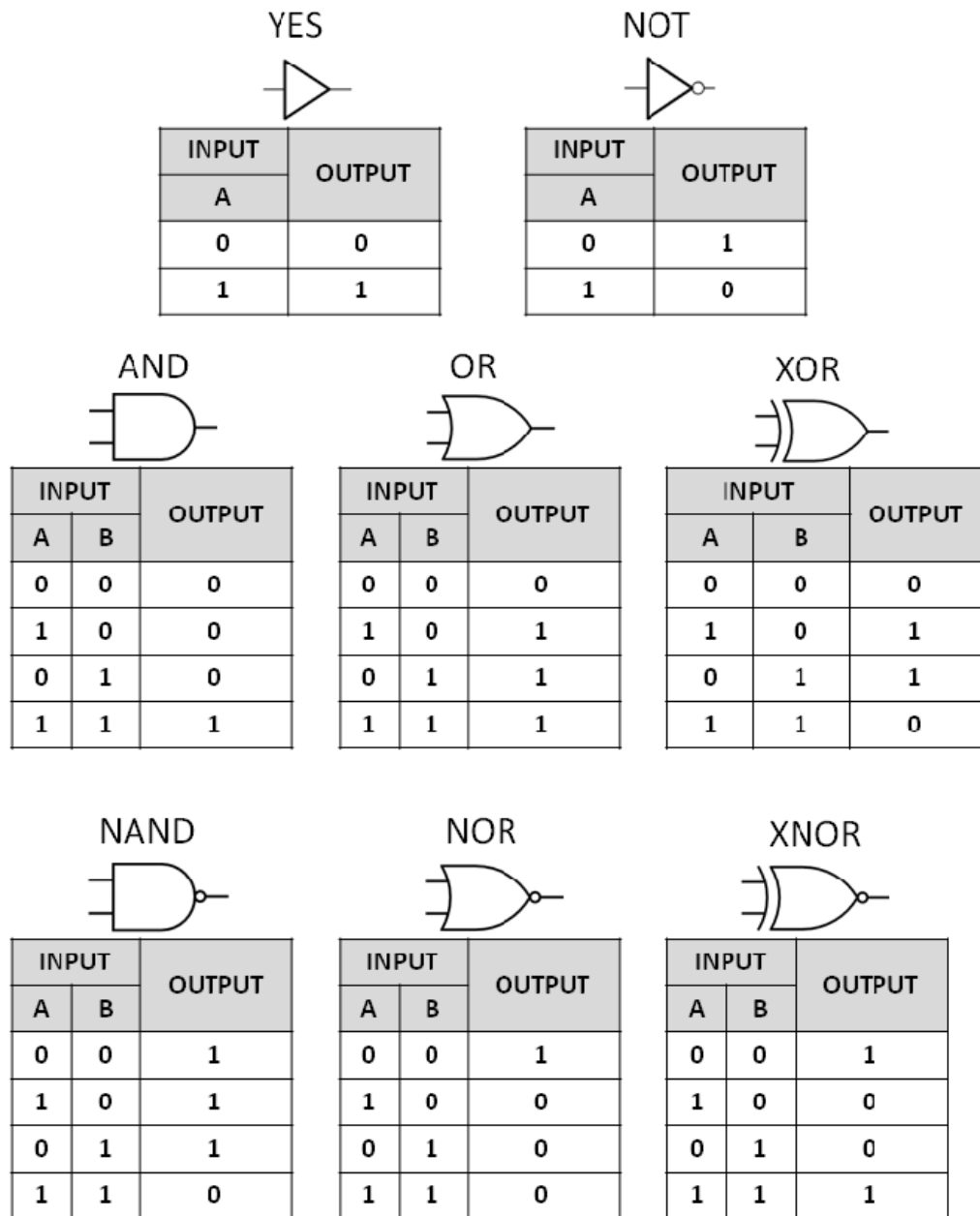


Figure 2: Classical Logic gates

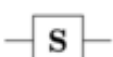
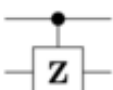
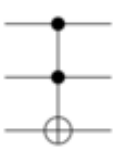
Operator	Gate(s)	Matrix
Pauli-X (X)	 	$\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$
Pauli-Y (Y)		$\begin{bmatrix} 0 & -i \\ i & 0 \end{bmatrix}$
Pauli-Z (Z)		$\begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$
Hadamard (H)		$\frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}$
Phase (S, P)		$\begin{bmatrix} 1 & 0 \\ 0 & i \end{bmatrix}$
$\pi/8$ (T)		$\begin{bmatrix} 1 & 0 \\ 0 & e^{i\pi/4} \end{bmatrix}$
Controlled Not (CNOT, CX)		$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{bmatrix}$
Controlled Z (CZ)	 	$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & -1 \end{bmatrix}$
SWAP	 	$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$
Toffoli (CCNOT, CCX, TOFF)		$\begin{bmatrix} 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \end{bmatrix}$

Figure 3: Quantum Logic gates

5 Quantum Noise(Photonic Qubits)

5.1 References

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5.2 Quantum Channel

Decoherence is defined as the loss of coherence. Dissipation is defined as the loss of energy of a system to the environment. We will provide a brief approach in showing how the motivation for defining an open quantum system comes. In studying real quantum systems, we can observe that no system is truly isolated. There is always an environment interacting with the system, and these interactions affect the system's dynamics. First, we recall that a closed quantum system's evolution is described by unitary operators and that no matter what the condition, a density operator will have hermiticity, positivity, and the trace condition. We start by considering the total Hilbert space of the system and its environment:

$$\mathcal{H}_{\text{tot}} = \mathcal{H}_S \otimes \mathcal{H}_E, \quad (26)$$

where $\mathcal{H}_S$ and $\mathcal{H}_E$ are the Hilbert spaces of the system and environment, respectively.

The total system evolves unitarily according to the Hamiltonian H_{tot} :

$$\rho_{\text{tot}}(t) = U(t) \rho_{\text{tot}}(0) U^\dagger(t), \quad U(t) = e^{-iH_{\text{tot}}t/\hbar}. \quad (27)$$

To describe the system alone, we trace out the environment degrees of freedom, obtaining the reduced density matrix:

$$\rho_S(t) = \text{Tr}_E[\rho_{\text{tot}}(t)]. \quad (28)$$

Assuming that initially the system and environment are uncorrelated, i.e., (Physically this means that the environment is interacting weakly at first)

$$\rho_{\text{tot}}(0) = \rho_S(0) \otimes \rho_E(0), \quad (29)$$

We can formally write the system evolution as

$$\rho_S(t) = \text{Tr}_E[U(t)(\rho_S(0) \otimes \rho_E(0))U^\dagger(t)]. \quad (30)$$

This evolution is generally non-unitary for the system, because tracing out the environment destroys coherence. To make the description more explicit, we can introduce a set of Kraus operators $\{K_i(t)\}$ acting only on the system, so that

$$\rho_S(t) = \sum_i K_i(t) \rho_S(0) K_i^\dagger(t), \quad \sum_i K_i^\dagger(t) K_i(t) = I. \quad (31)$$

The Kraus operators arise naturally by diagonalizing the initial environment state $\rho_E(0) = \sum_\alpha p_\alpha |\alpha\rangle\langle\alpha|$ and tracing over the environment basis $\{|\beta\rangle\}$:

$$K_i(t) = \sqrt{p_\alpha} \langle\beta|U(t)|\alpha\rangle, \quad (32)$$

where the sum over i accounts for all possible environment transitions.

Finally, if we assume the environment is memoryless (Markovian) and the system-environment coupling is weak, the evolution can be described continuously by the Lindblad master equation:

$$\frac{d\rho_S}{dt} = -\frac{i}{\hbar}[H_S, \rho_S] + \sum_j \left(L_j \rho_S L_j^\dagger - \frac{1}{2} \{L_j^\dagger L_j, \rho_S\} \right), \quad (33)$$

where H_S is the system Hamiltonian and L_j are Lindblad operators representing decoherence or dissipation.

This derivation motivates the study of open quantum systems: it shows how interactions with the environment lead to non-unitary evolution, decoherence, and dissipation, and provides a formalism (Kraus operators or Lindblad equation) to describe the system dynamics accurately.

In the quantum setting, the quantum system can be described by a density operator ρ acting on a Hilbert space. A quantum channel describes the most general physically allowed transformation of such states. Formally, a quantum channel is a linear map. Let S be the set of all possible density operators pertaining to a fixed physical system.

$$\mathcal{E} : S \rightarrow S$$

that satisfies the following properties:

- Complete Positivity: $\mathcal{E}$ is completely positive.
- Trace Preservation: $Tr(\mathcal{E}(\rho)) = Tr(\rho)$.

This map is surjective iff the evolution occurs via unitary operators. Maps satisfying these properties are called completely positive trace-preserving (CPTP) maps. (Observe that this ensures that the output is a valid density operator) The quantum channel $\mathcal{E}$ can be represented in the Kraus form

$$\mathcal{E}(\rho) = \sum_k E_k \rho E_k^\dagger$$

where the operators E_k satisfy the completeness condition

$$\sum_k E_k^\dagger E_k = I.$$

Intuitively, a quantum channel can be viewed as the quantum analogue of a classical noisy communication channel. In the classical case, the channel randomly maps input symbols to output symbols according to a probability distribution. In the quantum case, the channel acts on density matrices and may introduce decoherence, dissipation, or other forms of quantum noise. While classical channels transform probability distributions, quantum channels transform quantum states while preserving the fundamental physical constraints of quantum mechanics.

We will understand decoherence in more detail.

5.3 Pure Decoherence

This is also called dephasing. Recall the Bloch sphere aspect of a qubit, which I have discussed before.

So, pure decoherence destroys the relative phase, without altering the energy levels of the system.

Mathematically, consider a qubit,

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle.$$

By definition of density operator,

$$\rho = |\psi\rangle\langle\psi| = \begin{pmatrix} |\alpha|^2 & \alpha\beta^* \\ \alpha^*\beta & |\beta|^2 \end{pmatrix}.$$

Let after dephasing has started

$$\rho \equiv \rho(t) = \begin{pmatrix} |\alpha|^2 & \alpha\beta^*e^{-\gamma t} \\ \alpha^*\beta e^{-\gamma t} & |\beta|^2 \end{pmatrix}$$

where γ denotes the dephasing rate.

Temperature change, abrupt increase or decrease in the strength of magnetic fields can cause this.

Some examples of Quantum Noise

5.4 Bit-Flip Noise (X error)

Mathematically, Bit-flip noise is described by the Pauli-X operator. We have seen an intuitive understanding of what a quantum channel looks like:

$$\mathcal{E}(\rho) = \sum_k E_k \rho E_k^\dagger, \sum_k E_k^\dagger E_k = \mathbb{I}.$$

In case of bit-flips in a quantum channel with error probability p ,

$$\mathcal{E}(\rho) = (1-p)\rho + (pX\rho X)$$

In case of the photonic qubits, this causes an unintended rotation of the polarization state, for example, $|\rightarrow\rangle$ being measured as $|\uparrow\rangle$.

5.5 Phase-Flip Error Noise (Z error)

Mathematically, the phase flip error noise uses the Z to flip the sign of $|1\rangle$ to $-|1\rangle$.

$$Z \equiv \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$$

Here,

$$\mathcal{E}(\rho) = (1-p)\rho + pZ\rho Z$$

where p is the error probability.

This affects the relative phase of superpositions, which are essential for BB84 protocols, security.

5.6 Depolarizing Noise

Let p be the error probability.

$$\mathcal{E}(\rho) = (1-p)\rho + \frac{p}{3}(X\rho X + Y\rho Y + Z\rho Z)$$

Let me illustrate some concrete measurement examples. For

$$|\psi\rangle = \frac{1}{\sqrt{2}}[|0\rangle + |1\rangle]$$

$$\rho_{\text{initial}} = \begin{pmatrix} \frac{1}{2} & \frac{1}{2} \\ \frac{1}{2} & \frac{1}{2} \end{pmatrix}$$

Now, for pure decoherence, choose $e^{-\gamma t} = 0.5$ So,

$$\rho' = \begin{pmatrix} \frac{1}{2} & \frac{1}{4} \\ \frac{1}{4} & \frac{1}{2} \end{pmatrix}$$

For Bit-flip (choose $p = 0.1$)

$$\mathcal{E}(\rho) = 0.9 \begin{pmatrix} 0.5 & 0.5 \\ 0.5 & 0.5 \end{pmatrix} + 0.1 \begin{pmatrix} 0.5 & 0.5 \\ 0.5 & 0.5 \end{pmatrix} = \rho$$

For phase flip: ($p = 0.1$)

$$Z\rho Z = \begin{pmatrix} \frac{1}{2} & -\frac{1}{2} \\ -\frac{1}{2} & \frac{1}{2} \end{pmatrix}$$

$$\therefore \mathcal{E}(\rho) = 0.9 \begin{pmatrix} 0.5 & 0.5 \\ 0.5 & 0.5 \end{pmatrix} + 0.1 \begin{pmatrix} 0.5 & -0.5 \\ -0.5 & 0.5 \end{pmatrix} = \begin{pmatrix} 0.5 & 0.4 \\ 0.4 & 0.5 \end{pmatrix}$$

5.7 Measuring Decoherence

Recall,

$$D(\rho, \sigma) = \frac{1}{2} \text{tr}|\rho - \sigma|$$

The decoherence is measured, between ρ_0 and $\rho(t)$. Now, as decoherence increases, $D(\rho(t), \rho)$ increases.

Let $|\psi\rangle$ be a pure state, let $\rho(t)$ denote the altered state.

$$F(|\psi\rangle, \rho(t)) = \sqrt{\langle\psi|\rho|\psi\rangle}$$

Consider the concrete examples. Recall the $\rho(t) = \begin{pmatrix} \frac{1}{2} & \frac{1}{4} \\ \frac{1}{4} & \frac{1}{2} \end{pmatrix}$ So,

$$F^2 = \begin{pmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \end{pmatrix} \begin{pmatrix} \frac{1}{2} & \frac{1}{4} \\ \frac{1}{4} & \frac{1}{2} \end{pmatrix} \begin{pmatrix} \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} \end{pmatrix} = 0.75$$

indicates perfect state preservation.

5.8 How a pure state becomes a mixed state

Consider $|\psi\rangle = \frac{|0\rangle + |1\rangle}{\sqrt{2}}$. It is a superposition.

$$\rho = |\psi\rangle\langle\psi| = \begin{pmatrix} \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} \end{pmatrix} \begin{pmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \end{pmatrix} = \frac{1}{2} \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix};$$

$$\text{Tr}\rho^2 = 1.$$

Assume the action of a Z channel on ρ , and get

$$\rho' = (1-p)\rho + pZ\rho Z = \begin{pmatrix} \frac{1}{2} & \frac{1}{2}(1-2p) \\ \frac{1}{2}(1-2p) & \frac{1}{2} \end{pmatrix}$$

For $p = 0.5$

$$\rho' = \begin{pmatrix} \frac{1}{2} & 0 \\ 0 & \frac{1}{2} \end{pmatrix} \quad [\text{Off diagonal entries become 0}] \quad \text{--- (1)}$$

$$\rho'^2 = \begin{pmatrix} \frac{1}{2} & 0 \\ 0 & \frac{1}{2} \end{pmatrix} \begin{pmatrix} \frac{1}{2} & 0 \\ 0 & \frac{1}{2} \end{pmatrix} = \begin{pmatrix} \frac{1}{4} & 0 \\ 0 & \frac{1}{4} \end{pmatrix}$$

$$\therefore \text{Tr}(\rho')^2 = \frac{1}{2} < 1. \implies \rho' \text{ is a mixed state.}$$

We can see that the off-diagonal elements are correlated to coherence and superposition.

6 An Introduction to Core Machine Learning Techniques

Machine learning is best understood not as a single algorithmic trick, but as a collection of principles for extracting structure from data. At its heart lies a deceptively simple problem: given observations of the world, how should a system generalize beyond what it has seen? Different machine learning techniques answer this question in different ways, depending on assumptions about the data, the nature of the task, and the cost of being wrong.

This document introduces several foundational machine learning techniques, focusing on their conceptual motivations, mathematical structure, and practical roles. We emphasize supervised learning methods such as linear regression and classification, unsupervised techniques such as clustering, and ensemble methods that combine multiple learners to improve performance.

6.1 References

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3. James, G., Witten, D., Hastie, T., & Tibshirani, R. (2013). *An Introduction to Statistical Learning*. Springer.
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6.2 Supervised Learning: Linear Regression

Linear regression is often the first machine learning model encountered, not because it is trivial, but because it exposes the core learning problem in its simplest form.

6.2.1 Problem Setup

Given a dataset of input-output pairs

$$\{(x_i, y_i)\}_{i=1}^N,$$

where $x_i \in \mathbb{R}^d$ and $y_i \in \mathbb{R}$, linear regression assumes that the target variable can be approximated as a linear function of the inputs:

$$y \approx \mathbf{w}^\top \mathbf{x} + b.$$

The goal is to estimate the parameters $\mathbf{w}$ and b such that predictions closely match observed outputs.

6.2.2 Loss Function and Optimization

Learning proceeds by minimizing a loss function, typically the mean squared error:

$$L(\mathbf{w}, b) = \frac{1}{N} \sum_{i=1}^N (y_i - \mathbf{w}^\top \mathbf{x}_i - b)^2.$$

This choice is not arbitrary. Squared error corresponds to maximum likelihood estimation under a Gaussian noise assumption, revealing a deep connection between statistics and learning.

6.2.3 Interpretation and Limitations

Linear regression offers interpretability, convex optimization, and analytical solutions. However, it struggles when relationships are nonlinear, features are highly correlated, or noise distributions deviate significantly from Gaussian assumptions.

6.3 Classification Methods

While regression predicts continuous values, classification assigns inputs to discrete categories.

6.3.1 Linear Classification

In binary classification, labels take values $y \in \{0, 1\}$ or $y \in \{-1, 1\}$. Linear classifiers separate data using a hyperplane:

$$\mathbf{w}^\top \mathbf{x} + b = 0.$$

Examples include logistic regression and linear discriminant analysis. Logistic regression models the conditional probability

$$P(y = 1|\mathbf{x}) = \sigma(\mathbf{w}^\top \mathbf{x}),$$

where σ is the sigmoid function.

6.3.2 Decision Boundaries

Classification is fundamentally geometric. The model defines regions of input space corresponding to different labels. More complex classifiers induce nonlinear decision boundaries, often implicitly through feature transformations or kernel methods.

6.4 The k-Nearest Neighbors (k-NN) Algorithm

The k-nearest neighbors algorithm represents an extreme contrast to parametric models.

6.4.1 Core Idea

To classify or regress at a point $\mathbf{x}$, k-NN identifies the k closest training samples under a chosen distance metric and aggregates their labels or values.

There is no explicit training phase; the model is the dataset itself.

6.4.2 Bias-Variance Tradeoff

Small values of k lead to low bias but high variance, while large values smooth predictions but risk underfitting. This tradeoff mirrors fundamental statistical tensions present throughout machine learning.

6.4.3 Strengths and Weaknesses

k-NN is intuitive and flexible but computationally expensive at inference time and highly sensitive to feature scaling and irrelevant dimensions.

6.5 Unsupervised Learning and Clustering

Unsupervised learning seeks structure in unlabeled data. Clustering is among its most studied tasks.

6.5.1 k-Means Clustering

k-means partitions data into k clusters by minimizing within-cluster variance:

$$\sum_{j=1}^k \sum_{\mathbf{x}_i \in C_j} \|\mathbf{x}_i - \boldsymbol{\mu}_j\|^2,$$

where $\boldsymbol{\mu}_j$ is the centroid of cluster C_j .

The algorithm alternates between assignment and update steps, converging to a local minimum.

6.5.2 Choosing the Number of Clusters

Determining k is not trivial and motivates several diagnostic methods.

6.6 Model Selection in Clustering

6.6.1 Elbow Method

The elbow method plots clustering cost as a function of k . A sharp change in slope suggests diminishing returns beyond a certain number of clusters. While intuitive, it relies on subjective judgment.

6.6.2 Silhouette Method

The silhouette coefficient measures how well a point fits within its assigned cluster compared to others:

$$s = \frac{b - a}{\max(a, b)},$$

where a is intra-cluster distance and b is nearest-cluster distance. Values close to 1 indicate strong clustering structure.

6.6.3 Gap Statistic

The gap statistic compares observed clustering performance with that expected under a reference null distribution. It provides a more principled, statistical approach to selecting k .

6.7 Decision Trees

Decision trees model decisions through a hierarchy of splits on input features.

6.7.1 Structure and Learning

At each node, the algorithm selects a feature and threshold that maximizes information gain or reduces impurity (e.g., Gini index or entropy).

Trees recursively partition the feature space into axis-aligned regions.

6.7.2 Advantages and Drawbacks

Trees are interpretable and handle nonlinear relationships naturally. However, they are prone to overfitting and instability: small changes in data can yield very different trees.

6.8 Ensemble Methods

Ensemble methods reduce variance and bias by combining multiple models.

6.8.1 Bagging

Bootstrap aggregating, or bagging, trains multiple models on resampled versions of the dataset and averages their predictions. Random forests extend this idea by injecting feature randomness into decision trees.

6.8.2 Boosting

Boosting builds models sequentially, each focusing on correcting the errors of its predecessors. Algorithms such as AdaBoost and gradient boosting reinterpret learning as an additive optimization problem.

Boosting is powerful but sensitive to noise and requires careful regularization.

6.9 Neural Networks

In order to complement the physics-based simulations, a neural network model is employed as a function approximator to learn the relationship between system parameters and the resulting quantum bit error rate (QBER). Specifically, the network maps input features such as noise parameters and protocol characteristics to predicted QBER values.

The model used is a standard feedforward neural network consisting of multiple fully connected layers. Each layer applies a linear transformation followed by a nonlinear activation function, enabling the network to capture complex, nonlinear relationships present in the data. The depth and width of the network are chosen to balance expressive power with computational efficiency.

The network is trained using supervised learning, where the training dataset is generated from simulation results. The loss function used is the Mean Absolute Error (MAE), which measures the average deviation between predicted QBER values and those obtained from simulations. Minimizing this loss ensures that the model accurately approximates the underlying physical behavior.

To improve physical consistency, a physics-informed approach is incorporated in certain models. In this framework, additional structure or constraints derived from known physical principles are imposed during training. This helps the network generalize better and capture meaningful trends, rather than merely fitting the data.

Training is performed over the generated dataset, with performance evaluated on unseen data to assess generalization. The neural network is thus used not as a replacement for physical modeling, but as a complementary tool to efficiently approximate and analyze system behavior across different regimes.

6.10 Conclusion

Machine learning techniques form a connected ecosystem rather than isolated tools. Linear regression reveals the geometry of learning, k-NN exposes the role of locality, clustering uncovers latent structure, and ensembles demonstrate the power of collective inference.

Understanding these methods conceptually equips one to reason about new algorithms, diagnose failures, and adapt models to unfamiliar problems. The techniques presented here form a foundation

upon which more advanced methods—neural networks, kernel machines, and probabilistic models—are constructed.

Part II

Quantum Cryptography

7 Quantum Secure Communication

7.1 References

1. Katz, J., & Lindell, Y. (2020). *Introduction to Modern Cryptography*. CRC Press.
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7.2 A Brief History of Classical Cryptography Pre-RSA

The history of cryptography is a multi-millennial journey from linguistic intuition to rigorous mathematical proof, beginning with the manual era where security relied on Substitution and Transposition. The Caesar Cipher, defined by the shift function $E_k(p) = (p + k) \pmod{26}$, served as the foundational monoalphabetic substitution until the 9th-century mathematician Al-Kindi introduced Frequency Analysis. By recognizing that letters in natural language follow a non-uniform probability distribution, Al-Kindi turned codebreaking into a statistical science. This eventually necessitated the Vigenère Cipher, a polyalphabetic system defined as $c_i = (p_i + k_{i \pmod{m}}) \pmod{26}$. This "indecipherable" method stood for centuries until Friedrich Kasiski and Charles Babbage leveraged the mathematical repetitions of the keyword to break it in the mid-19th century. As the 20th century arrived, the speed of telecommunications demanded mechanization, leading to the era of Rotor Machines. The Enigma Machine epitomized this, using a series of rotating wired wheels to create a dynamic substitution alphabet that changed with every keystroke. Mathematically, the Enigma can be modeled as a composite permutation: $E = P \cdot R_1 \cdot R_2 \cdot R_3 \cdot T \cdot R_3^{-1} \cdot R_2^{-1} \cdot R_1^{-1} \cdot P^{-1}$, where P is the plugboard and T is the reflector. The breaking of this system by Alan Turing and the team at Bletchley Park marked the birth of computational cryptanalysis. This era was codified in 1949 by Claude Shannon, whose work defined Perfect Secrecy as $H(P|C) = H(P)$ and established the dual requirements of Confusion (masking the key-ciphertext relationship) and Diffusion (dissipating plaintext statistics across the ciphertext). The final pre-RSA chapter began in the 1970s with the transition to digital Block Ciphers. Horst Feistel developed a structured iterative approach where data is split into halves (L_i, R_i) and transformed via $R_{i+1} = L_i \oplus F(R_i, K_i)$. This "Feistel Network" became the heart of the Data Encryption Standard (DES), the first global cryptographic standard. However, these symmetric systems suffered from the "Key Distribution Problem"—the requirement that two parties share a secret key before communicating. This was finally solved in 1976 by Whitfield Diffie and Martin Hellman, who utilized the Discrete Logarithm Problem to allow for a secure key exchange over a public channel: $s = (g^b)^a \equiv (g^a)^b \pmod{p}$. This breakthrough in asymmetric logic provided the immediate theoretical framework for the emergence of the RSA algorithm just one year later.

7.3 RSA algorithm

The RSA algorithm is one of the popular means of ciphering and deciphering in classical cryptography. The security of RSA is not based on secret mechanical rotors or complex scrambling logic, but on the integer factorization problem—the fact that it is computationally easy to multiply two large prime numbers, but extremely difficult to factor the product back into the original primes. Unlike the symmetric ciphers that preceded it (like DES), RSA uses two different keys: a Public Key for encryption and a Private Key for decryption. Now I will describe the process for the Key generation. First two large prime

numbers are taken and their product(n) is taken. This " n " is used as the modulus for both the public and private keys. The totient $\phi(n) = (p - 1)(q - 1)$ is calculated. A public exponent e (often a small prime number co-prime to the totient) is chosen. The private exponent d is the modular multiplicative inverse of $e \pmod{\phi(n)}$, meaning $d \times e \equiv 1 \pmod{\phi(n)}$. Now I shall describe the encryption procedure. If a friend of mine wants to send me a message, he/she will use the public key (e, n) . The message has to be converted into an integer, M s.t $0 \leq M < n$. The ciphertext (C) is calculated using modular exponentiation:

$$C = M^e \pmod{n}$$

Even though the attacker knows e and n , they cannot easily reverse this operation to find M without knowing the private exponent d . To recover the original message, I have to use my Private Key (d, n) . I will then apply the private exponent to the ciphertext:

$$M = C^d \pmod{n}$$

Because of the relationship established during key generation ($d \times e \equiv 1 \pmod{\phi(n)}$), the math ensures that $(M^e)^d \pmod{n}$ returns precisely M . Now I will exhibit a simulation of the RSA algorithm in python.

```
import math

def rsa_simulation():
    print("---- RSA Algorithm Simulation ----")

    # 1. Key Generation
    # User-defined prime numbers
    p = int(input("Enter a prime number (p): "))
    q = int(input("Enter another prime number (q): "))

    # Compute the modulus n
    n = p * q
    print(f"Computed Modulus (n): {n}")

    # Calculate the totient phi(n)
    phi = (p - 1) * (q - 1)
    print(f"Computed Totient (phi): {phi}")

    # Choose public exponent e
    # e must be coprime to phi(n)
    e = int(input("Enter a small prime for public exponent (e, e.g., 65537): "))
    if math.gcd(e, phi) != 1:
        print("Error: e is not coprime to phi. Using default 65537 if possible.")
        e = 65537

    # Calculate private exponent d
    # d is the modular multiplicative inverse of e mod phi
    d = pow(e, -1, phi)

    print(f"\nPublic Key: (e={e}, n={n})")
    print(f"Private Key: (d={d}, n={n})")

    # 2. Encryption Process
    # Message M must be an integer where 0 <= M < n
    msg = int(input("\nEnter an integer message to encrypt (M < n): "))

    # C = M^e mod n
    ciphertext = pow(msg, e, n)
    print(f"Encrypted Ciphertext (C): {ciphertext}")

    # 3. Decryption Process
    # M = C^d mod n
    decrypted_msg = pow(ciphertext, d, n)
```

```

print(f"Decrypted Message (M): {decrypted_msg}")

# Verification
if msg == decrypted_msg:
    print("\nSuccess: The decrypted message matches the original.")
else:
    print("\nFailure: Decryption did not return the original message.")

if __name__ == "__main__":
    rsa.simulation()

```

7.4 How Quantum Algorithms challenge RSA

The primary challenge quantum computers pose to RSA is rooted in **Shor's Algorithm**, a quantum algorithm developed in 1994. In the classical landscape, the most efficient method for factoring an integer n is the General Number Field Sieve (GNFS), which has a sub-exponential but still prohibitive time complexity. Shor's Algorithm, however, leverages the principles of quantum superposition and entanglement to perform a quantum Fourier transform, finding the period of a modular function in polynomial time. Specifically, while a classical computer would take billions of years to factor a 2048-bit RSA modulus, a sufficiently powerful, fault-tolerant quantum computer could theoretically accomplish this in hours. This effectively collapses the "trapdoor" function of the modular exponentiation $C = M^e \pmod{n}$, as the quantum computer can derive the private exponent d directly from the public key by factoring n into its constituent primes p and q . I will discuss this in details in the upcoming chapters.

7.5 The Split in Modern Cryptography

Modern Cryptography has split into 2 broad branches. One branch uses methods of classical cryptography, to combat the weakness of quantum computers and quantum mechanics-based algorithms. These are called "Post-Quantum Cryptography". Some examples of this are

- Lattice-based Cryptography
- Elliptic Curve-based Cryptography
- Multivariate Quadratic Cryptography

I have just mentioned these to provide exposure. The other branch is called Quantum Key Distribution, which I will discuss in much detail.

7.6 QKD

In QKD, the information is encoded into quantum states (I will soon discuss the typical quantum states used). Usually the quantum states are taken as states characterized by different polarization angles. The above is done, since we are all familiar with light traveling through optical fibers or through air itself. When Alice sends a sequence of photons to Bob, they both choose to measure the states of the photon, w.r.t different mutually exclusive bases. (I will discuss these bases in detail when I do BB84 protocol). Now we are all familiar with how quantum measurement alters the states. So when Alice deploys a photon to Bob, we may expect a third party (say Eve) trying to spy on them. Now Eve knows about the bases, but she does not know which ones were used by Alice. So she tries a brute force method (intercepting the photon in between and measuring it with a random basis). This measurement changes the state. Alice and Bob can compare the no. of photons altered this way. The quantity QBER (Quantum Bit Error Rate) is used. A low QBER means they can safely start sharing information.

7.7 Physical means to implement this QKD

Developing the theory is a first step. But the most important is the implementation. The following methods try to tackle the implementation:

- **Photon Generation:** The information is encoded in photons. So we need some reliable means of generating them. The ideal case is a source generating single photons (the importance will come in the decoy state part). Quantum dots and nitrogen vacancy centers in diamonds are two ways. The most common are weak coherent pulses from lasers. (The process involves Poisson Distribution)

- **Photon Transmission:** Now that we know how to generate photons, we need some means of transmitting them or sending them. It can be done through optical fibers(The loss rate is high in this one). Another method uses free space to transmit them(Launched by China in August 2016, Micius was the first satellite to successfully demonstrate long-distance Quantum Key Distribution (QKD) and quantum entanglement distribution between space and Earth).
- **Photon Detection:** At the receiving end, Bob uses single photon detectors (take avalanche photodiode). The device clicks when a photon arrives but not perfectly. It is explored more deeply in the ML program.

7.8 Quantum No-Cloning Theorem

There exists no universal device capable of perfectly copying any arbitrary, unknown quantum state.

7.8.1 Universal Device

Take U as a 2-qubit unitary gate. Let $|\psi\rangle$ be an arbitrary unknown state. The operation of a cloning device would be:

$$U|\psi\rangle|0\rangle = |\psi\rangle|\psi\rangle$$

where $|0\rangle$ is an initial state.

7.9 Proof by Contradiction

7.9.1 Assumption

Let a unitary operator U exist such that:

$$U|\psi\rangle|0\rangle = |\psi\rangle|\psi\rangle \quad \text{for any } |\psi\rangle \quad (34)$$

Suppose this works for two different states $|\psi\rangle$ and $|\phi\rangle$:

1. $U|\psi\rangle|0\rangle = |\psi\rangle|\psi\rangle$
2. $U|\phi\rangle|0\rangle = |\phi\rangle|\phi\rangle$

7.9.2 Inner Product Analysis

Taking the inner product of the two resulting states:

$$(\langle\psi|\langle\psi|)(|\phi\rangle|\phi\rangle) = \langle\psi|\phi\rangle\langle\psi|\phi\rangle = \langle\psi|\phi\rangle^2$$

Now, looking at the left-hand side using the property of unitary operators ($U^\dagger U = I$):

$$\begin{aligned} (U|\psi\rangle|0\rangle)^\dagger (U|\phi\rangle|0\rangle) &= \langle 0|\langle\psi|U^\dagger U|\phi\rangle|0\rangle \\ &\Rightarrow \langle\psi|\phi\rangle\langle 0|0\rangle = \langle\psi|\phi\rangle \end{aligned}$$

7.9.3 Conclusion

Equating the two results:

$$\begin{aligned} \langle\psi|\phi\rangle &= \langle\psi|\phi\rangle^2 \\ \Rightarrow \langle\psi|\phi\rangle(\langle\psi|\phi\rangle - 1) &= 0 \end{aligned}$$

This equation holds true only if:

- $\langle\psi|\phi\rangle = 0$ (The states are **orthogonal**)
- OR $\langle\psi|\phi\rangle = 1$ (The states are **identical**)

Final Result: Since it is not true for *any* arbitrary $|\psi\rangle$ and $|\phi\rangle$, only two orthogonal states can be copied. Therefore, U is dependent on $|\psi\rangle$ and a universal cloning device is impossible.

Table 2: Summary Comparison: Classical vs. QKD

Feature	Classical (RSA/ECC)	Quantum Key Distribution (QKD)
Security Foundation	Mathematical Complexity	Laws of Physics
Eavesdropping	Undetectable	Immediately Detectable
Quantum Attack	Vulnerable (Shor's Algorithm)	Immune
Hardware	Standard Computers/Software	Dedicated Quantum Hardware
Key Type	Public/Private Key Pairs	Symmetric Random Keys

Table 3: Comparison of Security Foundations

Feature	Classical Cryptography (RSA, ECC)	Quantum Key Distribution (QKD)
Core Principle	Mathematical Hardness: Relies on the assumption that certain math problems (like factoring) are too hard to solve.	Laws of Physics: Relies on the fact that measuring a quantum system disturbs it.
Longevity	Temporary: Vulnerable to "Harvest Now, Decrypt Later" and future computing breakthroughs.	Future-Proof: Information-theoretically secure; immune to future increases in computing power.
Interception	Passive: An attacker can copy a key/message without the sender or receiver knowing.	Active: Any attempt to intercept the key introduces errors, alerting Alice and Bob.
Key Generation	Uses Pseudo-random number generators.	Uses True randomness from quantum processes.

7.10 Advantages of QKD over classical cryptography

7.11 Drawbacks of Quantum Key Distribution

Everything has its advantages and drawbacks. This section will explore the drawbacks of QKD. Given that a photon is moving through an optical fiber, it has been observed experimentally that energy loss occurs after 100km. A solution is to use free space as a medium, but the costs are high. The quantum hardware is not fully developed yet. While the physics provides unbreakable security, the equipment ironically is not up-to-the mark. QKD is merely a key exchange procedure, unlike the RSA, which could deal with keys and messages.

7.12 BB84 Protocol

7.12.1 Introduction to Quantum Cryptography

The latest line of cryptographical thinking is based on the principles of quantum entanglement and the inherent nature of quantum mechanics and measurements. While Shor's factorization algorithm is remarkably strong against classical RSA, quantum cryptography makes it possible to exchange a key on a public channel and perform the impossible task of detecting a third party with certainty.

7.12.2 Classical Binary Schematic

Before entering the quantum domain, we establish the binary foundation. The process is modeled as:

- **Alice (Encryption):** Combines the message with a random key.
- **Public Channel:** The scrambled message is transmitted.
- **Bob (Decryption):** Adds the secret key back to recover the message.

The binary rule of operation follows:

$$1 + 1 = 0, \quad 0 + 0 = 0, \quad 1 + 0 = 1, \quad 0 + 1 = 1$$

Remark: The safety of the channel depends critically on the secrecy and randomness of the key, which ensures the transmitted data remains inaccessible to a third party.

7.12.3 The Quantum Objective

In a classical channel, eavesdropping can occur without detection. Quantum cryptography modifies the objective: **Exchanging the key such that any third party is detected immediately.**

7.13 Main Principles of BB84

7.13.1 Bohr's Principle of Complementarity

Two observables are complementary if precise knowledge of one implies that all possible outcomes of measuring the other are equally probable. This prevents an eavesdropper from gaining full information about a photon's state.

7.13.2 Dual Channels of Communication

1. **Quantum Transmission Channel:** An array of polarized photons is sent from Alice to Bob.
2. **Classical Channel:** Information regarding the *preparation* and *measurement* bases is exchanged (but not the bit values).

7.14 Physical Implementation and Timing

7.14.1 Alice's Apparatus and Basis Encoding

Alice uses a source of horizontally polarized photons, modulated by a Pockel cell to produce four possible polarizations: $0^\circ, 45^\circ, 90^\circ, 135^\circ$. These are grouped into two bases:

- **Rectilinear ($\oplus$):** $\{| \rightarrow \rangle, | \uparrow \rangle\}$ corresponding to 0° and 90° .
- **Diagonal ($\otimes$):** $\{| \nearrow \rangle, | \nwarrow \rangle\}$ corresponding to 45° and 135° .

7.14.2 Timing and Noise Discrimination

Alice transmits photons at regular time intervals for three critical reasons:

- **Synchronization:** Detectors are opened only at specific intervals to correctly identify which photon corresponds to which bit.
- **Noise Discrimination:** Bob filters out "dark counts" (random false detections) occurring at odd times, improving the signal-to-noise ratio.
- **Security:** Irregular timing could allow an eavesdropper (Eve) to infer information via timing attacks. Regular intervals prevent side-channel leaks.

7.14.3 Bob's Apparatus

Bob uses a second Pockel cell to rotate the incoming photon's polarization by either 0° or 45° (his choice of basis), followed by a Polarizing Beam Splitter (PBS).

- If Bob chooses the **same basis** as Alice, the photons are perfectly correlated.
- If Bob chooses a **different basis**, the outcome is random (50% probability) due to complementarity.

7.15 Example and Sifting

7.15.1 Measurement Example

Consider Alice sending 9 photons with polarizations: $90^\circ, 90^\circ, 135^\circ, 0^\circ, 45^\circ, 135^\circ, 45^\circ, 45^\circ, 90^\circ$.

Alice's Bases: $\oplus, \oplus, \otimes, \oplus, \otimes, \otimes, \otimes, \otimes, \oplus$.

Bob's Bases: $\otimes, \oplus, \oplus, \otimes, \otimes, \otimes, \oplus, \oplus, \oplus$.

Bob's outcomes for those in the same basis are identical to Alice's. For mismatched bases, he receives random results (e.g., 45° or 135° for a 90° photon).

7.15.2 Sifting and The Random Key

Alice and Bob compare their bases over the public channel. They retain only the instances where their bases matched (e.g., 2, 5, 6, 8, 9) and discard the rest. These retained bits form the **Raw Key**.

7.16 Security and Practical Vulnerabilities

7.16.1 Passive Eavesdropping (The 25% Rule)

If Eve intercepts a 90° photon ($\oplus$ basis):

- **Case 1 (Correct Basis):** She measures in $\oplus$, gets 90° , and resends it. Bob detects no error.
- **Case 2 (Incorrect Basis):** She measures in $\otimes$, gets 45° or 135° . If she resends a 45° photon to Bob, Bob (measuring in $\oplus$) has a 50% chance of getting 0° instead of 90° .

This introduces a 25% Quantum Bit Error Rate (QBER), signaling the presence of a third party.

7.16.2 Practical Attack Vectors

- **No-Cloning Theorem:** Forbids Eve from making copies of a photon to measure later.
- **Photon-Number-Splitting (PNS):** If Alice sends > 1 photon per pulse, Eve can split one off. *Countermeasure: Decoy States.*
- **Detector Blinding:** Eve uses bright light to force detectors into classical mode. *Countermeasure: Better detector design or MDI-QKD.*
- **Trojan Horse:** Probing devices with light to extract internal states.

7.17 The Decoy-State Solution

In practice, attenuated lasers follow a Poisson distribution of photon numbers (0, 1, or ≥ 2 photons). Since multi-photon pulses are vulnerable to PNS attacks, the **Decoy-State Protocol** is implemented to detect such interventions by varying the intensity of the pulses. The Decoy State BB84 protocol introduces a defense mechanism. Alice sends pulses of varying intensities. Some pulses are signal states and others as decoy states.

Each pulse is randomly chosen to be:

- A **signal state** (with intensity μ , Mean Photon No. = 0.5)
- A **decoy state** with smaller intensity ν (≈ 0.1)
- Occasionally, a **vacuum state** (0 photons)

Remark: All states are indistinguishable to Eve.

7.18 Procedure of the Protocol

1. Preparation and Transmission

- Alice randomly picks both:
 - the quantum basis ($\times$ or $+$)
 - the pulse intensity
- She sends the corresponding weak coherent pulse to Bob.

2. Measurement

- Bob randomly chooses his measurement basis for each incoming photon and records the result.
- Later, he tells Alice which pulses he detected, and which bases he used (but not the results).

3. Sifting

- The overlapping bits are kept same.

- Alice then discloses which pulses were decoy and which were signal.

4. Parameter Estimation

- For each class (signal, decoy, vacuum), Alice and Bob compute:
 - The gain ($P(\text{Bob detecting a photon})$)
 - Quantum Bit Error Rate (QBER) (fraction of incorrect bits)
- By comparing these rates across intensities, they can estimate the fraction of detections that came from single-photon sources.

5. Security Analysis

- If Eve attempts a PNS (Photon Number Splitting) attack:

$$QBER(\text{signal state}) \neq QBER(\text{decoy state})$$

- From these discrepancies, Alice and Bob can upper bound Eve's information and lower bound the security key rate.

6. Key Generation

- They perform error correction and privacy amplification on the remaining bits, yielding a secure shared key.

7.19 Why it works

Eve cannot tell which pulse intensity she is dealing with. So, if she tampers, her interference alters the detection statistics inconsistently between signal and decoy state.

By analyzing these statistics, Alice and Bob can:

- Estimate the yields (detection probabilities) for single photon vs multiple photon pulses.
- Deviation can be detected.

7.20 Quantum Bit Error Rate Calculation in BB84 and Decoy-State BB84

The Quantum Bit Error Rate (QBER) quantifies the fraction of incorrect bits in the sifted key obtained after the basis reconciliation stage. It serves as a measure of the disturbance in the quantum channel and is used to detect noise or potential eavesdropping.

7.20.1 QBER in the BB84 Protocol

After the transmission and sifting stages, Alice and Bob possess correlated bit strings known as the sifted key.

To estimate the error rate, they publicly reveal a randomly selected subset of the sifted bits.

Let

$$N_{\text{error}}$$

denote the number of mismatched bits between Alice and Bob and

$$N_{\text{total}}$$

denote the total number of bits compared.

The Quantum Bit Error Rate is defined as

$$QBER = \frac{N_{\text{error}}}{N_{\text{total}}}$$

In experimental implementations using photon detectors, the QBER can also be expressed in terms of detector counts

$$QBER = \frac{C_{\text{wrong}}}{C_{\text{correct}} + C_{\text{wrong}}}$$

where C_{wrong} represents incorrect detection events and C_{correct} represents correct detection events.

If the QBER exceeds a predefined threshold (typically around 11% for ideal BB84 implementations), the protocol is aborted because the channel disturbance may indicate the presence of an eavesdropper.

7.20.2 QBER in the Decoy-State BB84 Protocol

In the decoy-state BB84 protocol, Alice randomly varies the intensity of the optical pulses. The pulses are classified as signal states and decoy states with different mean photon numbers.

The QBER is evaluated separately for each intensity class.

For a given intensity level μ , the error rate is defined as

$$E_{\mu} = \frac{N_{\text{error}}(\mu)}{N_{\text{detected}}(\mu)}$$

where

$$N_{\text{error}}(\mu)$$

denotes the number of erroneous detections associated with pulses of intensity μ , and

$$N_{\text{detected}}(\mu)$$

denotes the total number of detections for that intensity.

Using the decoy-state method, Alice and Bob estimate the error rate associated with single-photon pulses

$$e_1 = \frac{N_{\text{error}}^{(1)}}{N_{\text{detected}}^{(1)}}$$

where the superscript (1) refers to single-photon contributions.

This single-photon error rate is used in the security analysis and in determining the final secure key rate after error correction and privacy amplification.

7.21 Mean Photon Number

7.21.1 Basic Definitions

Let Alice use a phase-randomized, weak coherent source with a mean photon number μ . The probability $P(n)$ that n photons are emitted follows a Poisson distribution:

$$P(n) = e^{-\mu} \frac{\mu^n}{n!}$$

A coherent state $|\alpha\rangle$ is defined as:

$$|\alpha\rangle = e^{-|\alpha|^2/2} \sum_{n=0}^{\infty} \frac{\alpha^n}{\sqrt{n!}} |n\rangle$$

where $|\alpha|^2 = \mu$ (mean photon number).

- e_n denotes the error rate for an n -photon pulse ($e_0 = \frac{1}{2}$ for vacuum).
- Y_n (yield) := $P(\text{Bob detects} | \text{Alice sent } n\text{-photon pulse})$.

7.21.2 Significance of Mean Photon Number

- If μ is small, we are less likely to get a multi-photon pulse.
- As $\mu \approx 0$, the detection rate is reduced.

7.22 B-92 Protocol

We have seen that in the BB84 protocol, Alice can send binary information 0 and 1 through polarized photons in $\times$ or $+$ bases. Instead of 4 pairwise orthogonal bases, we just use $\{| \rightarrow \rangle, | \nearrow \rangle\}$ corresponding to 0 and 1 respectively.

7.22.1 How the B-92 Protocol Works

Bob, upon reviewing the photons, measures them in the randomly chosen basis $\otimes$ or $\oplus$. If Alice sends photons only in states $|\rightarrow\rangle$ and $|\nearrow\rangle$ to Bob, then Bob can determine the $|\rightarrow\rangle$ state unambiguously only in those cases where he happens to choose the $\{\lvert\uparrow\rangle, \lvert\leftrightarrow\rangle\}$ basis and the outcome is $\lvert\uparrow\rangle$. Only when he finds the received photon in the $\lvert\uparrow\rangle$ state, he can be certain that it is not of the $|\nearrow\rangle$ state as the two states are mutually orthogonal and by default it should be in the $|\rightarrow\rangle$ state. For $|\nearrow\rangle$ state we proceed similarly. In all other cases, he can be certain that the incoming photon is not in the state.

7.22.2 B-92 Protocol in Steps

- Alice sends a stream of bits, in such a way that the bit '0' is encoded with a polarized photon in state $|\rightarrow\rangle$ and 1 in $|\nearrow\rangle$.
- Bob assigns the value '0' to his bit if he chooses the basis $\{\lvert\nearrow\rangle, \lvert\nwarrow\rangle\}$ and 1 if $\{\lvert\uparrow\rangle, \lvert\rightarrow\rangle\}$. Thus he forms a random string of bit.
- Bob then measures the polarization of the received photon in the chosen basis. Based on the outcome of the measurements Bob forms another string of bits, which he calls the control bits.
- Bob assigns 0 for $|\rightarrow\rangle$ or $|\nearrow\rangle$, 1 for $|\nwarrow\rangle$ or $|\uparrow\rangle$.

7.22.3 Key Generation and Example

At this point Alice has a stream of bits with $0 \leftrightarrow |\rightarrow\rangle$ and $1 \leftrightarrow |\nearrow\rangle$, whereas Bob has 2 strings. A stream of bits with $0 \leftrightarrow \{\lvert\nearrow\rangle, \lvert\nwarrow\rangle\}$ and $1 \leftrightarrow \{\lvert\uparrow\rangle, \lvert\rightarrow\rangle\}$ that he chooses, as well as a stream of control bits $0 \leftrightarrow |\rightarrow\rangle$ or $|\nearrow\rangle$ and $1 \leftrightarrow |\nwarrow\rangle$ or $|\uparrow\rangle$. Bob informs Alice the values of the control bits without telling her the bases that he used. They decide to keep only those values of the bits, for which the values of the control bits are one. These values of the bits form the key.

Case 1: $\oplus$ basis 1 value is assigned to his bit. The polarization of the photon is $|\rightarrow\rangle$. Bob's outcome is definitely going to be $|\rightarrow\rangle$ and 0 is assigned for the control bit. *How can Bob conclude that Alice sent him the photon in the state $|\rightarrow\rangle$?* Ans: He cannot. As if Alice had sent a 1 via the $|\nearrow\rangle$ state, then due to the decomposition:

$$|\nearrow\rangle = \frac{1}{\sqrt{2}}(|\rightarrow\rangle + |\uparrow\rangle)$$

there is a 50% chance that the outcome will also be $|\rightarrow\rangle$. So, a detection in the state $|\rightarrow\rangle$ means Alice could have sent $|\rightarrow\rangle$ or $|\nearrow\rangle$.

Case 2: $\otimes$ basis Bob assigns 0 to his bit.

$$|\rightarrow\rangle = \frac{1}{\sqrt{2}}(|\uparrow\rangle - |\nwarrow\rangle)$$

There is a 50% chance that his outcome will be $|\uparrow\rangle$ or $|\nwarrow\rangle$. If it is $|\nwarrow\rangle$, 0 is assigned to his control bit. Here Bob can be sure about one thing: Alice did not send $|\nearrow\rangle$ because it is orthogonal to $|\nwarrow\rangle$. And Bob knows the states Alice used ($|\rightarrow\rangle, |\nearrow\rangle$), so he knows with certainty that Alice sent $|\rightarrow\rangle$.

7.22.4 Security and Eavesdropping

Eve has a 50/50 chance of choosing the correct basis and incorrect basis. The presence of Eve is inferred if Eve's measurements lead to a wrong outcome for Bob. This can happen in two ways:

1. Alice sends a 0 in $|\rightarrow\rangle$. Bob detects it in $|\uparrow\rangle$ and concludes Eve sent "1" with $|\nwarrow\rangle$. This happens when Eve and Bob in $\oplus$. $P(D) = 12.5\%$.
2. Alice sends 1 in $|\nearrow\rangle$, wrongly concluding Alice sent a 0 in $|\rightarrow\rangle$. Eve has used $\oplus$.

QC relies on the statistical certainty that attackers' luck runs out.

7.22.5 Conditions for B-92

Let Alice send two non-orthogonal pure states $|\psi_0\rangle$ and $|\psi_1\rangle$. Let $s(\text{overlap}) := |\langle\psi_0|\psi_1\rangle|$, clearly $s \in (0, 1)$.

- **Unambiguous State Discrimination:** Eve can sometimes identify the state without error, but with inconclusive outcomes. The max prob of $P(USD) = 1 - s$.
- **Minimum Error (Helstrom) Measurement:** Eve minimizes her average probability of making a wrong guess.

$$P_e = \frac{1}{2}(1 - \sqrt{1 - s^2}), \quad P_H = 1 - P_e$$

For N transmitted signals, if Eve needs to avoid creating any detectable discrepancy, $Pr(\text{all correct}) = P^N$.

7.22.6 Advantage of B-92 over BB-84

1. Requires less hardware. Simplifies state preparation and measurement. Reduces calibration errors that arise from switching between two bases.
2. Easier State Preparation: Only 2 states are used. The source does not need to rapidly switch between phase settings. This makes B-92 more robust against imperfection in photon generation.
3. Lower Classical Communication Overhead: BB84 requires both Alice and Bob to announce and compare bases publicly to discard mismatched measurements. B92 sidesteps that (via control bit).
4. Security through state non-orthogonality. Non-orthogonal states cannot be distinguished.

But: B-92 is more sensitive to channel noise and photon loss. B-92 is vulnerable to PNS attack even more than BB84.

7.23 Two Way B-92 Protocol

The two-way B92 protocol is an extension of the original B92 quantum key distribution scheme proposed by Charles Bennett. In the original protocol, Alice sends one of two non-orthogonal quantum states to Bob. Because these states are non-orthogonal, they cannot be perfectly distinguished, which guarantees security against eavesdropping.

The two-way version introduces a round-trip structure in which a quantum state travels from Alice to Bob and then back to Alice. This additional interaction allows improved verification of disturbances in the quantum channel and can increase robustness against certain attacks.

7.23.1 Choice of Quantum States

The protocol uses two non-orthogonal states to encode classical bits. A convenient representation uses polarization states or qubit states on the Bloch sphere.

Let the computational basis be

$$\begin{aligned} |0\rangle &= \begin{pmatrix} 1 \\ 0 \end{pmatrix}, & |1\rangle &= \begin{pmatrix} 0 \\ 1 \end{pmatrix} \\ |\psi_0\rangle &= |0\rangle \\ |\psi_1\rangle &= |+\rangle = \frac{|0\rangle + |1\rangle}{\sqrt{2}} \end{aligned}$$

The inner product between the two states is:

$$\langle\psi_0|\psi_1\rangle = \frac{1}{\sqrt{2}}$$

Alice encodes bits using two non-orthogonal states

$$|\psi_0\rangle = |0\rangle$$

$$|\psi_1\rangle = |+\rangle = \frac{|0\rangle + |1\rangle}{\sqrt{2}}$$

The inner product between the two states is:

$$\langle\psi_0|\psi_1\rangle = \frac{1}{\sqrt{2}}$$

Since the states are not orthogonal, perfect discrimination is impossible.

7.23.2 Preparation of the Initial State

Alice begins by generating a random classical bit string. For each bit she prepares a qubit according to the encoding rule

$$\begin{aligned} [0 &\rightarrow |\psi_0\rangle] \\ [1 &\rightarrow |\psi_1\rangle] \end{aligned}$$

The prepared qubit is then transmitted to Bob through the quantum channel.

7.23.3 Bob's Measurement and Encoding Operation

Upon receiving the qubit, Bob does not immediately measure it. Instead, he performs one of two operations depending on a randomly chosen bit.

If Bob chooses bit (0), he applies the identity operator

$$I = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$

If Bob chooses bit (1), he applies a unitary operation such as a phase flip

$$[Z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}]$$

After performing the chosen operation, Bob sends the qubit back to Alice.

7.23.4 Alice's Measurement

Once the qubit returns, Alice measures it in a basis that allows her to distinguish whether Bob applied the identity operation or the phase flip.

Because the original states are non-orthogonal, Alice sometimes obtains inconclusive results. Only the conclusive outcomes are retained for key generation.

7.23.5 Sifting Procedure

Alice and Bob communicate over a classical authenticated channel to identify which rounds produced valid detection events. Inconclusive measurements are discarded. The remaining bits form the sifted key.

7.23.6 Quantum Bit Error Rate Estimation

To detect the presence of noise or an eavesdropper, Alice and Bob randomly select a subset of the sifted key and publicly compare their bits.

The quantum bit error rate is calculated as

$$QBER = \frac{N_{\text{error}}}{N_{\text{tested}}}$$

If the QBER exceeds a predetermined threshold, the key is discarded.

7.23.7 Key Distillation

If the observed error rate is below the security threshold, Alice and Bob perform classical post-processing steps including error correction and privacy amplification. These steps remove residual errors and reduce any information that a potential eavesdropper might have gained.

The resulting bit string constitutes the final shared secret key.

7.24 Differential Phase B-92 Protocol

The DPS-B92 protocol is a hybrid quantum key distribution scheme that combines ideas from the Differential Phase Shift (DPS) protocol and the B92 protocol. Information is encoded in the *relative phase* between consecutive coherent pulses rather than in single isolated qubits. Security arises from the non-orthogonality of the phase states and the impossibility of perfectly distinguishing them.

7.24.1 State Preparation by Alice

Alice prepares a sequence of weak coherent pulses using a laser source. Each pulse has mean photon number $\mu \ll 1$ so that the probability of multi-photon emission is minimized.

Information is encoded in the **phase difference** between consecutive pulses.

Alice randomly chooses one of two relative phases

$$\Delta\phi \in \{0, \pi\}$$

These correspond to two non-orthogonal signal states:

$$|\psi_0\rangle = |\alpha\rangle_k |\alpha\rangle_{k+1}$$

$$|\psi_1\rangle = |\alpha\rangle_k |-\alpha\rangle_{k+1}$$

where $|\alpha\rangle$ denotes a coherent state.

The encoding rule is

$$0 \rightarrow \Delta\phi = 0, \quad 1 \rightarrow \Delta\phi = \pi$$

Thus the key information is stored in the **phase difference between adjacent pulses**.

7.24.2 Transmission Through the Quantum Channel

Alice sends the pulse train through a quantum channel (typically optical fiber or free space).

Channel noise, photon loss, and environmental interaction may introduce decoherence and phase errors.

The received state at Bob's side can be written as

$$\rho_{\text{Bob}} = \mathcal{E}(\rho_{\text{Alice}})$$

where $\mathcal{E}$ represents the quantum channel.

7.24.3 Measurement by Bob

Bob uses a **Mach-Zehnder interferometer** with a one-pulse delay.

The interferometer combines pulse k and pulse $k+1$ so that their phase difference determines which detector clicks.

Two detectors are used:

$$D_0, \quad D_1$$

Detection outcomes correspond to the phase difference

$$\Delta\phi = 0 \rightarrow D_0$$

$$\Delta\phi = \pi \rightarrow D_1$$

If neither detector clicks (vacuum event), the result is discarded.

7.24.4 Sifting Stage

Bob publicly announces the time slots where a detection occurred.

Alice keeps only those pulses corresponding to Bob's valid detections.

Both parties now share a raw key sequence.

7.24.5 Error Estimation

To estimate the channel error rate, Alice and Bob randomly choose a subset of the raw key and reveal it publicly.

This sample is used to compute the Quantum Bit Error Rate (QBER).

7.24.6 Quantum Bit Error Rate (QBER)

The Quantum Bit Error Rate measures the fraction of mismatched bits between Alice and Bob.

If

$$N_{\text{error}}$$

is the number of mismatched bits and

$$N_{\text{total}}$$

is the total number of compared bits, then

$$\text{QBER} = \frac{N_{\text{error}}}{N_{\text{total}}}$$

In DPS-B92 systems, QBER can also be expressed in terms of detector counts:

$$\text{QBER} = \frac{C_{\text{wrong}}}{C_{\text{correct}} + C_{\text{wrong}}}$$

where

$$C_{\text{correct}}$$

is the number of clicks corresponding to the correct phase difference and

$$C_{\text{wrong}}$$

represents clicks at the incorrect detector.

A high QBER indicates possible eavesdropping or excessive channel noise.

7.24.7 Error Correction and Privacy Amplification

If the QBER is below a security threshold, Alice and Bob perform classical post-processing:

Error correction removes discrepancies in the raw key.

Privacy amplification reduces any information potentially obtained by an eavesdropper.

The final result is a shared secret key.

7.25 Entanglement-based B-92 protocol

The entanglement-based B92 protocol is a variant of the original B92 quantum key distribution scheme in which non-orthogonal signal states are generated through measurements on entangled particles. A source distributes entangled qubit pairs between Alice and Bob. Alice's measurement collapses Bob's particle into one of two non-orthogonal states which encode the key information.

7.25.1 Entangled State Preparation

A source prepares maximally entangled qubit pairs in the Bell state

$$|\Phi^+\rangle = \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$$

One qubit of each pair is sent to Alice and the other to Bob through a quantum channel.

7.25.2 Alice's Measurement and State Preparation

Alice randomly chooses one of two measurement bases corresponding to two non-orthogonal states. Her measurement projects Bob's qubit into a corresponding conditional state.

For example, Alice uses the states

$$|\psi_0\rangle = |0\rangle$$

$$|\psi_1\rangle = \cos\theta|0\rangle + \sin\theta|1\rangle$$

where $0 < \theta < \frac{\pi}{2}$.

After Alice measures her qubit, Bob's qubit collapses into one of these states depending on the outcome.

7.25.3 Transmission Through the Quantum Channel

Bob receives his particle through a noisy quantum channel which can be represented by a completely positive trace preserving map

$$\rho'_B = \mathcal{E}(\rho_B)$$

Noise or decoherence may modify the state and introduce errors.

7.25.4 Bob's Measurement

Bob performs a measurement designed to distinguish the two non-orthogonal states probabilistically.

He uses two projective measurements

$$P_0 = I - |\psi_1\rangle\langle\psi_1|$$

$$P_1 = I - |\psi_0\rangle\langle\psi_0|$$

If P_0 clicks, Bob infers bit 0.

If P_1 clicks, Bob infers bit 1.

If neither detector clicks, the result is inconclusive and discarded.

7.25.5 Sifting Procedure

Bob publicly announces the time slots in which he obtained a conclusive measurement outcome.

Alice keeps only those corresponding bits from her sequence.

The remaining bits form the raw key.

7.25.6 Quantum Bit Error Rate (QBER)

To estimate the security of the channel, Alice and Bob reveal a randomly chosen subset of the raw key.

Let

$$N_{\text{error}}$$

denote the number of mismatched bits between Alice and Bob and

$$N_{\text{total}}$$

denote the total number of compared bits.

The Quantum Bit Error Rate is defined as

$$QBER = \frac{N_{\text{error}}}{N_{\text{total}}}$$

In experimental implementations it can also be written using detector counts

$$QBER = \frac{C_{\text{wrong}}}{C_{\text{correct}} + C_{\text{wrong}}}$$

where C_{wrong} represents incorrect detection events and C_{correct} represents correct detections.

7.25.7 Post-Processing

If the QBER is below a predefined threshold, Alice and Bob perform classical post-processing steps including error correction and privacy amplification. These procedures remove residual errors and reduce any information potentially obtained by an eavesdropper, resulting in a secure shared secret key.

7.26 State Rotation Protocol B92 (SRP-B92)

The State Rotation Protocol B92 (SRP-B92) is a modification of the standard B92 quantum key distribution scheme in which Alice applies a controlled unitary rotation to the quantum states before transmission. The protocol uses non-orthogonal states whose relative angle is determined by a rotation parameter. This approach improves resistance to certain eavesdropping strategies while preserving the simplicity of the B92 framework.

7.26.1 State Preparation

Alice prepares a sequence of qubits encoding random classical bits. Two non-orthogonal quantum states are used to represent the logical bits

$$|\psi_0\rangle = |0\rangle$$

$$|\psi_1\rangle = \cos\theta|0\rangle + \sin\theta|1\rangle$$

where $0 < \theta < \frac{\pi}{2}$ determines the degree of non-orthogonality.

7.26.2 State Rotation

Before transmission, Alice applies a unitary rotation operator

$$R(\phi) = \begin{pmatrix} \cos\phi & -\sin\phi \\ \sin\phi & \cos\phi \end{pmatrix}$$

The transmitted states become

$$|\psi'_0\rangle = R(\phi)|\psi_0\rangle$$

$$|\psi'_1\rangle = R(\phi)|\psi_1\rangle$$

The rotation angle ϕ is known to Bob but unknown to potential eavesdroppers.

7.26.3 Transmission Through the Quantum Channel

Alice sends the rotated states through a quantum channel. Noise and environmental interaction may modify the states. The channel action can be represented as

$$\rho' = \mathcal{E}(\rho)$$

where $\mathcal{E}$ denotes a completely positive trace preserving quantum operation.

7.26.4 Bob's Measurement

Bob applies the inverse rotation $R(-\phi)$ to the received state in order to restore the original basis.

$$|\tilde{\psi}\rangle = R(-\phi)|\psi'\rangle$$

Bob then performs measurements designed to distinguish the two non-orthogonal states. The measurement operators are

$$P_0 = I - |\psi_1\rangle\langle\psi_1|$$

$$P_1 = I - |\psi_0\rangle\langle\psi_0|$$

If P_0 clicks Bob infers bit 0. If P_1 clicks Bob infers bit 1. If neither clicks the result is inconclusive and discarded.

7.26.5 Sifting Procedure

Bob announces the time slots in which he obtained a conclusive measurement outcome. Alice keeps only the corresponding bits from her original sequence. These remaining bits form the raw key.

7.26.6 Quantum Bit Error Rate (QBER)

To estimate the noise level of the channel, Alice and Bob publicly compare a randomly selected subset of the raw key.

Let

$$N_{\text{error}}$$

be the number of mismatched bits between Alice and Bob and

$$N_{\text{total}}$$

be the number of revealed bits.

The Quantum Bit Error Rate is defined as

$$QBER = \frac{N_{\text{error}}}{N_{\text{total}}}$$

Experimentally it can also be expressed in terms of detector counts

$$QBER = \frac{C_{\text{wrong}}}{C_{\text{correct}} + C_{\text{wrong}}}$$

where C_{wrong} denotes incorrect detection events and C_{correct} denotes correct detection events.

7.26.7 Post Processing

If the QBER is below a predefined security threshold, Alice and Bob perform classical post-processing. Error correction removes discrepancies in the raw key, and privacy amplification reduces any information that may have been obtained by an eavesdropper. The final output is a shared secret cryptographic key.

7.27 SARG04 Protocol

The SARG04 protocol is a quantum key distribution scheme that uses the same four quantum states as the BB84 protocol but modifies the classical sifting procedure. The protocol encodes classical information using non-orthogonal state sets rather than orthogonal bases, which provides improved resistance to certain eavesdropping strategies in practical implementations using weak coherent pulses.

7.27.1 State Preparation

Alice prepares a sequence of qubits randomly chosen from the four states

$$|0\rangle, \quad |1\rangle, \quad |+\rangle, \quad |-\rangle$$

where

$$|+\rangle = \frac{|0\rangle + |1\rangle}{\sqrt{2}}$$

$$|-\rangle = \frac{|0\rangle - |1\rangle}{\sqrt{2}}$$

These correspond to two conjugate bases.

7.27.2 Encoding Rule

Instead of associating bits directly with bases as in BB84, the SARG04 protocol encodes bits using pairs of non-orthogonal states.

Alice randomly chooses one of the following state pairs

$$\{|0\rangle, |+\rangle\}, \quad \{|0\rangle, |-\rangle\}, \quad \{|1\rangle, |+\rangle\}, \quad \{|1\rangle, |-\rangle\}$$

Within each pair, one state represents bit 0 and the other represents bit 1.

7.27.3 Transmission Through the Quantum Channel

Alice sends the prepared qubits to Bob through a quantum channel. The channel may introduce noise, losses, or decoherence. The action of the channel on the state can be represented as

$$\rho' = \mathcal{E}(\rho)$$

where $\mathcal{E}$ is a completely positive trace preserving map.

7.27.4 Bob's Measurement

Bob randomly chooses one of two measurement bases:

$$Z = \{|0\rangle, |1\rangle\}$$

$$X = \{|+\rangle, |-\rangle\}$$

He measures each incoming qubit and records the outcome.

7.27.5 Sifting Procedure

After the transmission phase, Alice publicly announces the pair of states that contained the transmitted state but does not reveal which one was actually sent.

Bob checks whether his measurement outcome is compatible with only one state in the announced pair.

If this condition is satisfied, Bob keeps the corresponding bit. Otherwise, the result is discarded.

7.27.6 Quantum Bit Error Rate (QBER)

To estimate the error rate of the quantum channel, Alice and Bob publicly compare a randomly selected subset of their sifted key.

Let

$$N_{\text{error}}$$

denote the number of mismatched bits between Alice and Bob and

$$N_{\text{total}}$$

denote the total number of compared bits.

The Quantum Bit Error Rate is defined as

$$QBER = \frac{N_{\text{error}}}{N_{\text{total}}}$$

In experimental implementations the QBER may also be expressed in terms of detector counts

$$QBER = \frac{C_{\text{wrong}}}{C_{\text{correct}} + C_{\text{wrong}}}$$

where C_{wrong} denotes incorrect detections and C_{correct} denotes correct detections.

7.27.7 Post Processing

If the QBER is below the security threshold, Alice and Bob perform classical post-processing procedures. Error correction reconciles discrepancies between their keys, and privacy amplification reduces any information potentially obtained by an eavesdropper, producing the final secure key.

7.28 Decoy State B-92

The modifications are similar to the ones we did for the Decoy State BB84 protocol.

Table 4: Comparison between BB84 and B92 Protocols

Feature	BB84 Protocol	B92 Protocol
Hardware & Complexity	Requires more hardware for state preparation and measurement.	Requires less hardware; simplifies state preparation and measurement.
State Preparation	Involves more complex switching between bases.	Easier; uses only 2 non-orthogonal states and doesn't require rapid phase setting switches.
Classical Overhead	Higher; requires Alice and Bob to publicly announce and compare bases.	Lower; sidesteps base comparison via the use of "control bits".
Robustness	Generally more robust against certain physical imperfections.	More robust against photon generation imperfections but highly sensitive to channel noise and loss.
Security Vulnerability	Standard vulnerability to various attacks.	More vulnerable to Photon Number Splitting (PNS) attacks than BB84.
Security Basis	Relies on the use of two conjugate bases.	Relies on the non-orthogonality of the two chosen states.

7.29 Quantum Bit Error Rate Calculation in B92 and Decoy-State B92

The Quantum Bit Error Rate (QBER) quantifies the fraction of incorrect bits in the sifted key obtained after the measurement and sifting stages. It provides an estimate of the disturbance introduced by noise in the quantum channel or by a potential eavesdropper.

7.29.1 QBER in the B92 Protocol

In the B92 protocol, Alice transmits two non-orthogonal quantum states representing classical bits. Bob performs measurements designed to probabilistically distinguish these states.

Only conclusive detection events are retained during the sifting stage, while inconclusive outcomes are discarded.

To estimate the QBER, Alice and Bob publicly compare a randomly chosen subset of the sifted key. Let

$$N_{\text{error}}$$

denote the number of mismatched bits between Alice and Bob and

$$N_{\text{total}}$$

denote the total number of compared bits.

The Quantum Bit Error Rate is defined as

$$QBER = \frac{N_{\text{error}}}{N_{\text{total}}}$$

In practical detector-based implementations, the QBER can also be expressed using count statistics

$$QBER = \frac{C_{\text{wrong}}}{C_{\text{correct}} + C_{\text{wrong}}}$$

where C_{wrong} denotes the number of incorrect detection events and C_{correct} denotes the number of correct detections.

If the measured QBER exceeds the security threshold, the protocol is aborted.

7.29.2 QBER in the Decoy-State B92 Protocol

In the decoy-state B92 protocol, Alice varies the intensity of the transmitted pulses, producing signal states and decoy states with different mean photon numbers.

The error rate is evaluated separately for each intensity level.

For a pulse intensity μ , the observed error rate is

$$E_{\mu} = \frac{N_{\text{error}}(\mu)}{N_{\text{detected}}(\mu)}$$

where

$$N_{\text{error}}(\mu)$$

represents the number of erroneous detections associated with pulses of intensity μ , and

$$N_{\text{detected}}(\mu)$$

represents the total number of detection events for that intensity.

Using the decoy-state method, Alice and Bob estimate the error rate associated with single-photon pulses

$$e_1 = \frac{N_{\text{error}}^{(1)}}{N_{\text{detected}}^{(1)}}$$

where the superscript (1) denotes contributions from single-photon states.

This single-photon error rate plays a central role in the security analysis and in

8 QKD Simulations

8.1 Interactive Decoherence Simulation

Keeping in mind the theory behind decoherence, we try to simulate an interactive Python program, which highlights the physics. Now the code does the following:

- The plus basis vector is chosen as the initial condition.
- The Pauli Matrices X, Y, and Z are used to define the quantum noises discussed in the decoherence chapter.
- The first parameter is taken as the noise probability p, which attempts to replicate the error rates in quantum hardware. This allows us to control the extent of the noise damping happening.
- The second parameter is the number of steps. It allows us to replicate the noise channel over and over, thus establishing the completion of one cycle.
- The first quantity that we are measuring is $|\rho_{01}|$, which as we can recall indicates superposition.
- The second quantity that we are measuring is

$$\text{Tr}(\rho^2)$$

.

8.2 Interactive Eve Simulation

The code does the following things:

- The quantum noises discussed are simulated.
- Raw key bits are generated.
- Eve's attack is executed via the random function.
- Both the original QBER and altered QBER rate is plotted.

The codes can be found at [simulations](#).

Part III

Application

9 Machine Learning Application

9.1 Problem Statement

The objective of this work is to model the effect of quantum noise on photonic qubits using machine learning.

Given a qubit state:

$$|\psi\rangle = a|0\rangle + b|1\rangle, \quad a, b \in \mathbb{C}, \quad |a|^2 + |b|^2 = 1$$

and a noise parameter $p \in [0, 1]$, the system evolves under a quantum channel as:

$$\rho' = (1 - p)\rho + pX\rho X$$

Learning Task:

$$(p, \psi) \longrightarrow \rho'$$

The goal is to learn this transformation from data and accurately predict the noisy quantum state.

9.2 Methodology

Dataset Generation:

- Random qubits $|\psi\rangle = a|0\rangle + b|1\rangle$, with $a, b \in \mathbb{C}$
- Normalization: $|a|^2 + |b|^2 = 1$
- Noise parameter $p \sim \mathcal{U}(0, 1)$
- Apply Pauli-X channel:

$$\rho' = (1 - p)\rho + pX\rho X$$

Feature–Target Mapping:

- Input: $(p, \text{Re}(a), \text{Im}(a), \text{Re}(b), \text{Im}(b))$
- Output: Flattened density matrix $\rho' \in \mathbb{C}^{2 \times 2}$

Models Used:

- Lasso and Ridge Regression
- Gradient Boosting Regressor
- Neural Networks: Standard, Residual

Evaluation:

$$\text{MSE} = \frac{1}{N} \sum_i \|\rho_i - \hat{\rho}_i\|_F^2$$

with Monte Carlo 5 runs.

9.3 Experimental Model Specifications

Linear and Ensemble Baselines: The classical ML architectures are given as follows:

- **Lasso Regression:** Multi-output model with L1 penalty ($\alpha = 0.001$).
- **Ridge Regression:** Multi-output model with L2 penalty ($\alpha = 1.0$).
- **Gradient Boosting:** Ensemble of 100 trees with a maximum depth of 3.

The Deep Learning architectures are given as follows:

- **Standard NN:** Fully connected MLP ($5 \rightarrow 32 \rightarrow 32 \rightarrow 8$) using ReLU activations.
- **Residual NN:** Includes a $5 \rightarrow 32$ projection layer followed by two residual blocks and a $32 \rightarrow 8$ output layer.
- **Residual Block:** Employs a skip connection defined as $y = x + f(x)$, where $f(x)$ consists of two linear layers with an internal ReLU.

9.4 Training Strategy

- **Optimizer:** Adam algorithm with a learning rate of 10^{-3} .
- **Loss Function:** Mean Squared Error (MSE) on flattened density matrix elements.
- **Evaluation:** 200 training epochs per split across 5 Monte Carlo iterations.

9.5 Sampling Statistics and Physical Structure

I. Noise Parameter Distribution (p)

- Sampling protocol:

$$p \sim \mathcal{U}(0, 1)$$

- Physical interpretation: Bit-flip probability within the Pauli- X quantum channel.
- Observation: Histogram analysis confirms approximately uniform coverage across the interval $[0, 1]$, ensuring unbiased sampling from identity evolution to complete bit-flip action.

II. State Coefficient Statistics (a, b)

- Complex amplitudes generated via normalized Gaussian sampling:

$$|a|^2 + |b|^2 = 1$$

- Real and imaginary components remain approximately symmetric about zero:

$$\mu \approx 0$$

- Distribution corresponds to randomized pure-state sampling over the qubit state manifold.

9.6 Model Performance

The models are evaluated using the **Matrix Frobenius Norm Mean Squared Error (MSE)**, calculated after inverse transformation of scaled predictions:

$$\text{MSE} = \frac{1}{N} \sum_{i=1}^N \|\rho_{\text{true},i} - \rho_{\text{pred},i}\|_F^2$$

Model Architecture	Mean MSE	Standard Deviation (σ)
Lasso Regression	0.281084	± 0.001478
Ridge Regression	0.281142	± 0.001502
Gradient Boosting (GB)	0.104926	± 0.003149
Standard Neural Network	0.056171	± 0.009152
Residual NN	0.011213	± 0.000912

Table 5: Aggregated performance across 5-fold Monte Carlo splits.

9.7 Model Analysis

9.7.1 Linear Model Insufficiency (Lasso/Ridge)

- The mapping from state amplitudes to density matrix elements is governed by the outer product $\rho = |\psi\rangle\langle\psi|$.
- This transformation is fundamentally quadratic; linear regressors are structurally incapable of capturing these non-linear dependencies, leading to an MSE ceiling of ≈ 0.28 .

9.7.2 Residual NN Structural Advantage

- The Pauli-X channel can be expressed as: $\rho_{noisy} = \rho + p(X\rho X - \rho)$.
- This identifies the output as the original state plus a noise-dependent perturbation.
- The Residual architecture ($y = x + \mathcal{F}(x)$) provides gradient stability, thereby allowing the model to learn the differential noise effect rather than reconstructing the full density matrix.

9.7.3 Comparative Stability

- Standard NNs and GB models lack explicit skip connections, requiring higher complexity to approximate the identity component of the noise map.
- The Residual NN demonstrates the lowest variance ($\sigma = 0.000912$), indicating superior optimization convergence and generalization across Monte Carlo splits.

9.8 Code

This code can be accessed at [repository](#).

9.9 Future Work and Limitations

Here we have chosen the toy X model and used simulated data. The long-term goal is to incorporate all forms of Quantum Noise and use ML to predict a noisy baseline QBER.

A Appendix

The following section explores the TLS noise observed in superconducting qubits and its modeling via the advanced Deep Learning architecture.